

36 Flexible static memory controller (FSMC)

This section applies to the whole STM32F40x/41x family only.

36.1 FSMC main features

The FSMC block is able to interface with synchronous and asynchronous memories and 16-bit PC memory cards. Its main purpose is to:

- Translate the AHB transactions into the appropriate external device protocol
- Meet the access timing requirements of the external devices

All external memories share the addresses, data and control signals with the controller. Each external device is accessed by means of a unique chip select. The FSMC performs only one access at a time to an external device.

The FSMC has the following main features:

- Interfaces with static memory-mapped devices including:
 - Static random access memory (SRAM)
 - NOR/OneNAND flash memory
 - PSRAM (4 memory banks)
- Two banks of NAND flash with ECC hardware that checks up to 8 Kbytes of data
- 16-bit PC Card compatible devices
- Supports burst mode access to synchronous devices (NOR flash and PSRAM)
- 8- or 16-bit wide databus
- Independent chip select control for each memory bank
- Independent configuration for each memory bank
- Programmable timings to support a wide range of devices, in particular:
 - Programmable wait states (up to 15)
 - Programmable bus turnaround cycles (up to 15)
 - Programmable output enable and write enable delays (up to 15)
 - Independent read and write timings and protocol, so as to support the widest variety of memories and timings
- Write enable and byte lane select outputs for use with PSRAM and SRAM devices
- Translation of 32-bit wide AHB transactions into consecutive 16-bit or 8-bit accesses to external 16-bit or 8-bit devices
- A Write FIFO, 2-word long (16-word long for STM32F42x and STM32F43x), each word is 32 bits wide, only stores data and not the address. Therefore, this FIFO only buffers AHB write burst transactions. This makes it possible to write to slow memories and free the AHB quickly for other operations. Only one burst at a time is buffered: if a new AHB burst or single transaction occurs while an operation is in progress, the FIFO is drained. The FSMC inserts wait states until the current memory access is complete.
- External asynchronous wait control

The FSMC registers that define the external device type and associated characteristics are usually set at boot time and do not change until the next reset or power-up. However, it is possible to change the settings at any time.

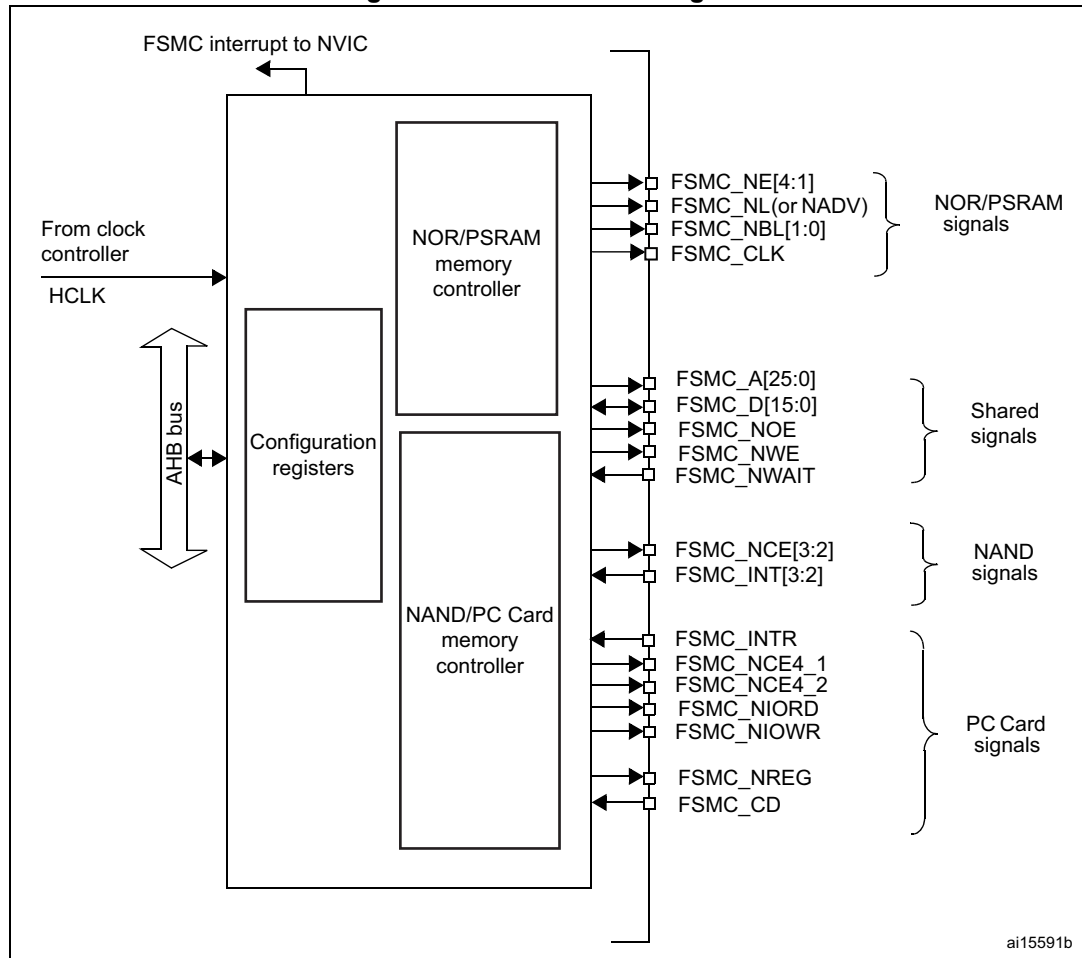
36.2 Block diagram

The FSMC consists of four main blocks:

- The AHB interface (including the FSMC configuration registers)
- The NOR flash/PSRAM controller
- The NAND flash/PC Card controller
- The external device interface

The block diagram is shown in [Figure 434](#).

Figure 434. FSMC block diagram



36.3 AHB interface

The AHB slave interface enables internal CPUs and other bus master peripherals to access the external static memories.

AHB transactions are translated into the external device protocol. In particular, if the selected external memory is 16 or 8 bits wide, 32-bit wide transactions on the AHB are split into consecutive 16- or 8-bit accesses. The FSMC Chip Select (FSMC_NEx) does not

toggle between consecutive accesses except when performing accesses in mode D with the extended mode enabled.

The FSMC generates an AHB error in the following conditions:

- When reading or writing to an FSMC bank which is not enabled
- When reading or writing to the NOR flash bank while the FACCEN bit is reset in the FSMC_BCRx register.
- When reading or writing to the PC Card banks while the input pin FSMC_CD (Card Presence Detection) is low.

The effect of this AHB error depends on the AHB master which has attempted the R/W access:

- If it is the Cortex[®]-M4 with FPU CPU, a hard fault interrupt is generated
- If is a DMA, a DMA transfer error is generated and the corresponding DMA channel is automatically disabled.

The AHB clock (HCLK) is the reference clock for the FSMC.

36.3.1 Supported memories and transactions

General transaction rules

The requested AHB transaction data size can be 8-, 16- or 32-bit wide whereas the accessed external device has a fixed data width. This may lead to inconsistent transfers.

Therefore, some simple transaction rules must be followed:

- AHB transaction size and memory data size are equal
There is no issue in this case.
- AHB transaction size is greater than the memory size
In this case, the FSMC splits the AHB transaction into smaller consecutive memory accesses in order to meet the external data width.
- AHB transaction size is smaller than the memory size
Asynchronous transfers may or not be consistent depending on the type of external device.
 - Asynchronous accesses to devices that have the byte select feature (SRAM, ROM, PSRAM).
 - a) FSMC allows write transactions accessing the right data through its byte lanes NBL[1:0]
 - b) Read transactions are allowed. All memory bytes are read and the useless ones are discarded. The NBL[1:0] are kept low during read transactions.
 - Asynchronous accesses to devices that do not have the byte select feature (NOR and NAND flash 16-bit).
This situation occurs when a byte access is requested to a 16-bit wide flash memory. Clearly, the device cannot be accessed in byte mode (only 16-bit words can be read from/written to the flash memory) therefore:
 - a) Write transactions are not allowed
 - b) Read transactions are allowed. All memory bytes are read and the useless ones are discarded. The NBL[1:0] are set to 0 during read transactions.

Configuration registers

The FSMC can be configured using a register set. See [Section 36.5.6](#), for a detailed description of the NOR flash/PSRAM control registers. See [Section 36.6.8](#), for a detailed description of the NAND flash/PC Card registers.

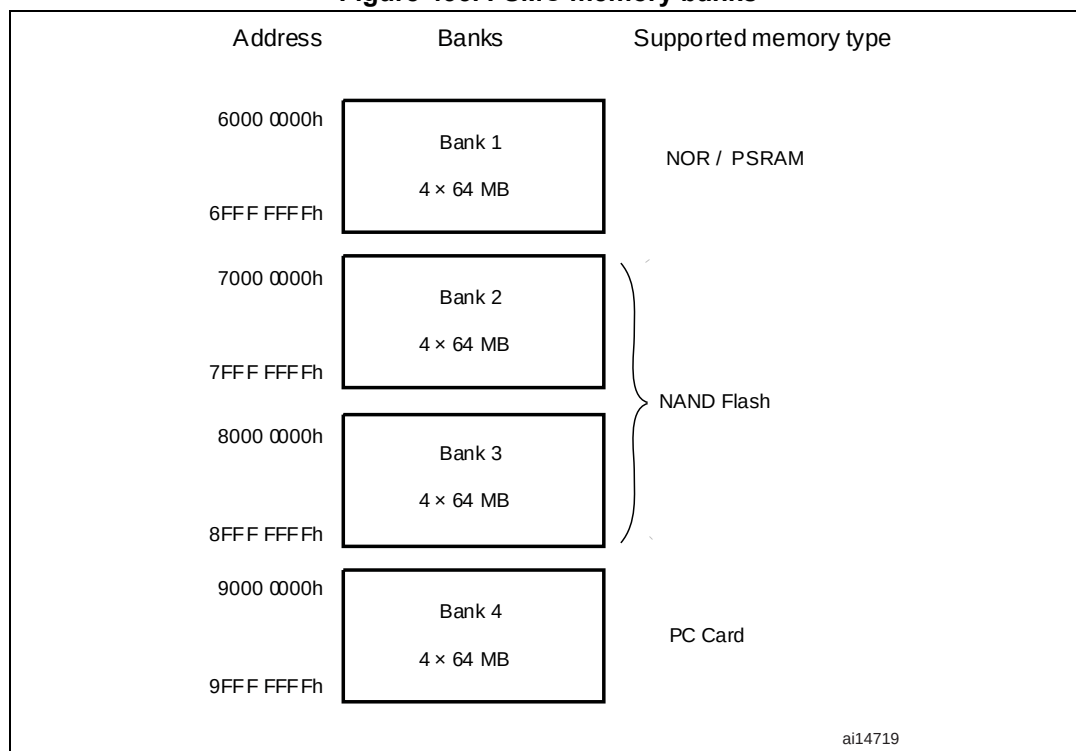
36.4 External device address mapping

From the FSMC point of view, the external memory is divided into 4 fixed-size banks of 256 Mbytes each (Refer to [Figure 435](#)):

- Bank 1 used to address up to four NOR flash or PSRAM memory devices. This bank is split into 4 NOR/PSRAM subbanks with four dedicated chip selects, as follows:
 - Bank 1 - NOR/PSRAM 1
 - Bank 1 - NOR/PSRAM 2
 - Bank 1 - NOR/PSRAM 3
 - Bank 1 - NOR/PSRAM 4
- Banks 2 and 3 used to address NAND flash devices (1 device per bank)
- Bank 4 used to address a PC Card device

For each bank the type of memory to be used is user-defined in the Configuration register.

Figure 435. FSMC memory banks



36.4.1 NOR/PSRAM address mapping

HADDR[27:26] bits are used to select one of the four memory banks as shown in [Table 216](#).

Table 216. NOR/PSRAM bank selection

HADDR[27:26] ⁽¹⁾	Selected bank
00	Bank 1 - NOR/PSRAM 1
01	Bank 1 - NOR/PSRAM 2
10	Bank 1 - NOR/PSRAM 3
11	Bank 1 - NOR/PSRAM 4

1. HADDR are internal AHB address lines that are translated to external memory.

HADDR[25:0] contain the external memory address. Since HADDR is a byte address whereas the memory is addressed in words, the address actually issued to the memory varies according to the memory data width, as shown in the following table.

Table 217. External memory address

Memory width ⁽¹⁾	Data address issued to the memory	Maximum memory capacity (bits)
8-bit	HADDR[25:0]	64 Mbyte x 8 = 512 Mbit
16-bit	HADDR[25:1] >> 1	64 Mbyte/2 x 16 = 512 Mbit

1. In case of a 16-bit external memory width, the FSMC internally uses HADDR[25:1] to generate the address for external memory FSMC_A[24:0].
Whatever the external memory width (16-bit or 8-bit), FSMC_A[0] should be connected to external memory address A[0].

Wrap support for NOR flash/PSRAM

Wrap burst mode for synchronous memories is not supported. The memories must be configured in linear burst mode of undefined length.

36.4.2 NAND/PC Card address mapping

In this case, three banks are available, each of them divided into memory spaces as indicated in [Table 218](#).

Table 218. Memory mapping and timing registers

Start address	End address	FSMC Bank	Memory space	Timing register
0x9C00 0000	0x9FFF FFFF	Bank 4 - PC card	I/O	FSMC_PIO4 (0xB0)
0x9800 0000	0x9BFF FFFF		Attribute	FSMC_PATT4 (0xAC)
0x9000 0000	0x93FF FFFF		Common	FSMC_PMEM4 (0xA8)
0x8800 0000	0x8BFF FFFF	Bank 3 - NAND flash	Attribute	FSMC_PATT3 (0x8C)
0x8000 0000	0x83FF FFFF		Common	FSMC_PMEM3 (0x88)
0x7800 0000	0x7BFF FFFF	Bank 2 - NAND flash	Attribute	FSMC_PATT2 (0x6C)
0x7000 0000	0x73FF FFFF		Common	FSMC_PMEM2 (0x68)

For NAND flash memory, the common and attribute memory spaces are subdivided into three sections (see in [Table 219](#) below) located in the lower 256 Kbytes:

- Data section (first 64 Kbytes in the common/attribute memory space)
- Command section (second 64 Kbytes in the common / attribute memory space)
- Address section (next 128 Kbytes in the common / attribute memory space)

Table 219. NAND bank selections

Section name	HADDR[17:16]	Address range
Address section	1X	0x020000-0x03FFFF
Command section	01	0x010000-0x01FFFF
Data section	00	0x000000-0x0FFFFF

The application software uses the 3 sections to access the NAND flash memory:

- **To send a command to NAND flash memory:** the software must write the command value to any memory location in the command section.
- **To specify the NAND flash address that must be read or written:** the software must write the address value to any memory location in the address section. Since an address can be 4 or 5 bytes long (depending on the actual memory size), several consecutive writes to the address section are needed to specify the full address.
- **To read or write data:** the software reads or writes the data value from or to any memory location in the data section.

Since the NAND flash memory automatically increments addresses, there is no need to increment the address of the data section to access consecutive memory locations.

36.5 NOR flash/PSRAM controller

The FSMC generates the appropriate signal timings to drive the following types of memories:

- Asynchronous SRAM and ROM
 - 8-bit
 - 16-bit
 - 32-bit
- PSRAM (Cellular RAM)
 - Asynchronous mode
 - Burst mode for synchronous accesses
 - Multiplexed or nonmultiplexed
- NOR flash
 - Asynchronous mode
 - Burst mode for synchronous accesses
 - Multiplexed or nonmultiplexed

The FSMC outputs a unique chip select signal NE[4:1] per bank. All the other signals (addresses, data and control) are shared.

For synchronous accesses, the FSMC issues the clock (CLK) to the selected external device only during the read/write transactions. This clock is a submultiple of the HCLK clock. The size of each bank is fixed and equal to 64 Mbytes.

Each bank is configured by means of dedicated registers (see [Section 36.5.6](#)).

The programmable memory parameters include access timings (see [Table 220](#)) and support for wait management (for PSRAM and NOR flash accessed in burst mode).

Table 220. Programmable NOR/PSRAM access parameters

Parameter	Function	Access mode	Unit	Min.	Max.
Address setup	Duration of the address setup phase	Asynchronous	AHB clock cycle (HCLK)	0	15
Address hold	Duration of the address hold phase	Asynchronous, muxed I/Os	AHB clock cycle (HCLK)	1	15
Data setup	Duration of the data setup phase	Asynchronous	AHB clock cycle (HCLK)	1	256
Bus turn	Duration of the bus turnaround phase	Asynchronous and synchronous read/write	AHB clock cycle (HCLK)	0	15
Clock divide ratio	Number of AHB clock cycles (HCLK) to build one memory clock cycle (CLK)	Synchronous	AHB clock cycle (HCLK)	2	16
Data latency	Number of clock cycles to issue to the memory before the first data of the burst	Synchronous	Memory clock cycle (CLK)	2	17

36.5.1 External memory interface signals

[Table 221](#), [Table 222](#) and [Table 223](#) list the signals that are typically used to interface NOR flash, SRAM and PSRAM.

Note: Prefix “N”. specifies the associated signal as active low.

NOR flash, nonmultiplexed I/Os

Table 221. Nonmultiplexed I/O NOR flash

FSMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:0]	O	Address bus
D[15:0]	I/O	Bidirectional data bus
NE[x]	O	Chip select, x = 1..4
NOE	O	Output enable
NWE	O	Write enable
NL(=NADV)	O	Latch enable (this signal is called address valid, NADV, by some NOR flash devices)
NWAIT	I	NOR flash wait input signal to the FSMC

NOR flash memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

NOR flash, multiplexed I/Os

Table 222. Multiplexed I/O NOR flash

FSMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:16]	O	Address bus
AD[15:0]	I/O	16-bit multiplexed, bidirectional address/data bus
NE[x]	O	Chip select, x = 1..4
NOE	O	Output enable
NWE	O	Write enable
NL(=NADV)	O	Latch enable (this signal is called address valid, NADV, by some NOR flash devices)
NWAIT	I	NOR flash wait input signal to the FSMC

NOR-flash memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

PSRAM/SRAM, nonmultiplexed I/Os

Table 223. Nonmultiplexed I/Os PSRAM/SRAM

FSMC signal name	I/O	Function
CLK	O	Clock (only for PSRAM synchronous access)
A[25:0]	O	Address bus
D[15:0]	I/O	Data bidirectional bus
NE[x]	O	Chip select, x = 1..4 (called NCE by PSRAM (Cellular RAM i.e. CRAM))
NOE	O	Output enable
NWE	O	Write enable
NL(= NADV)	O	Address valid only for PSRAM input (memory signal name: NADV)
NWAIT	I	PSRAM wait input signal to the FSMC
NBL[1]	O	Upper byte enable (memory signal name: NUB)
NBL[0]	O	Lowest byte enable (memory signal name: NLB)

PSRAM memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

PSRAM, multiplexed I/Os

Table 224. Multiplexed I/O PSRAM

FSMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:16]	O	Address bus
AD[15:0]	I/O	16-bit multiplexed, bidirectional address/data bus
NE[x]	O	Chip select, x = 1..4 (called NCE by PSRAM (Cellular RAM i.e. CDRAM))
NOE	O	Output enable
NWE	O	Write enable
NL(= NADV)	O	Address valid PSRAM input (memory signal name: NADV)
NWAIT	I	PSRAM wait input signal to the FSMC
NBL[1]	O	Upper byte enable (memory signal name: NUB)
NBL[0]	O	Lower byte enable (memory signal name: NLB)

PSRAM memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

36.5.2 Supported memories and transactions

[Table 225](#) below displays an example of the supported devices, access modes and transactions when the memory data bus is 16-bit for NOR, PSRAM and SRAM. Transactions not allowed (or not supported) by the FSMC in this example appear in gray.

Table 225. NOR flash/PSRAM controller: example of supported memories and transactions

Device	Mode	R/W	AHB data size	Memory data size	Allowed/ not allowed	Comments
NOR flash (muxed I/Os and nonmuxed I/Os)	Asynchronous	R	8	16	Y	-
	Asynchronous	W	8	16	N	-
	Asynchronous	R	16	16	Y	-
	Asynchronous	W	16	16	Y	-
	Asynchronous	R	32	16	Y	Split into two FSMC accesses
	Asynchronous	W	32	16	Y	Split into two FSMC accesses
	Asynchronous page	R	-	16	N	Mode is not supported
	Synchronous	R	8	16	N	-
	Synchronous	R	16	16	Y	-
	Synchronous	R	32	16	Y	-

Table 225. NOR flash/PSRAM controller: example of supported memories and transactions

Device	Mode	R/W	AHB data size	Memory data size	Allowed/ not allowed	Comments
PSRAM (multiplexed and nonmultiplexed I/Os)	Asynchronous	R	8	16	Y	-
	Asynchronous	W	8	16	Y	Use of byte lanes NBL[1:0]
	Asynchronous	R	16	16	Y	-
	Asynchronous	W	16	16	Y	-
	Asynchronous	R	32	16	Y	Split into two FSMC accesses
	Asynchronous	W	32	16	Y	Split into two FSMC accesses
	Asynchronous page	R	-	16	N	Mode is not supported
	Synchronous	R	8	16	N	-
	Synchronous	R	16	16	Y	-
	Synchronous	R	32	16	Y	-
	Synchronous	W	8	16	Y	Use of byte lanes NBL[1:0]
	Synchronous	W	16 / 32	16	Y	-
SRAM and ROM	Asynchronous	R	8 / 16	16	Y	-
	Asynchronous	W	8 / 16	16	Y	Use of byte lanes NBL[1:0]
	Asynchronous	R	32	16	Y	Split into two FSMC accesses
	Asynchronous	W	32	16	Y	Split into two FSMC accesses. Use of byte lanes NBL[1:0]

36.5.3 General timing rules

Signals synchronization

- All controller output signals change on the rising edge of the internal clock (HCLK)
- In synchronous mode (read or write), all output signals change on the rising edge of HCLK. Whatever the CLKDIV value, all outputs change as follows:
 - NOEL/NWEL/ NEL/NADV L/ NADV H/ NBLL/ Address valid outputs change on the falling edge of FSMC_CLK clock.
 - NOEH/ NWEH / NEH/ NOEH/NBLH/ Address invalid outputs change on the rising edge of FSMC_CLK clock.

36.5.4 NOR flash/PSRAM controller asynchronous transactions

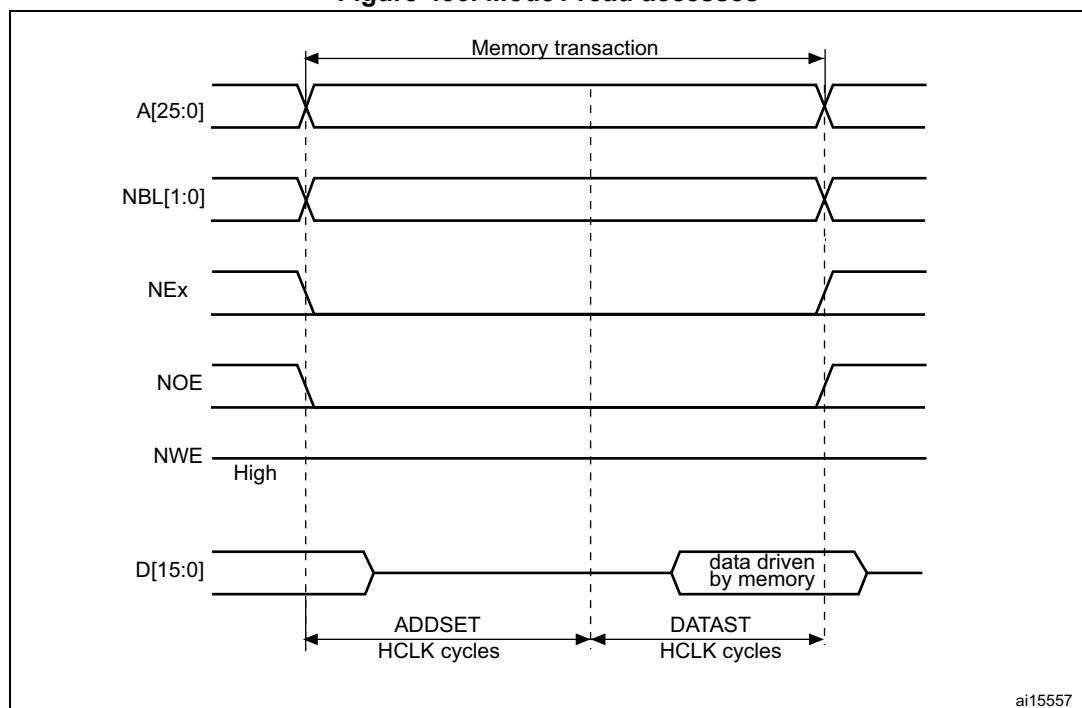
Asynchronous static memories (NOR flash memory, PSRAM, SRAM)

- Signals are synchronized by the internal clock HCLK. This clock is not issued to the memory
- The FSMC always samples the data before de-asserting the NOE signals. This guarantees that the memory data-hold timing constraint is met (chip enable high to data transition, usually 0 ns min.)
- If the extended mode is enabled (EXTMOD bit is set in the FSMC_BCRx register), up to four extended modes (A, B, C and D) are available. It is possible to mix A, B, C and D modes for read and write operations. For example, read operation can be performed in mode A and write in mode B.
- If the extended mode is disabled (EXTMOD bit is reset in the FSMC_BCRx register), the FSMC can operate in Mode1 or Mode2 as follows:
 - Mode 1 is the default mode when SRAM/PSRAM memory type is selected (MTYP[0:1] = 0x0 or 0x01 in the FSMC_BCRx register)
 - Mode 2 is the default mode when NOR memory type is selected (MTYP[0:1] = 0x10 in the FSMC_BCRx register).

Mode 1 - SRAM/PSRAM (CRAM)

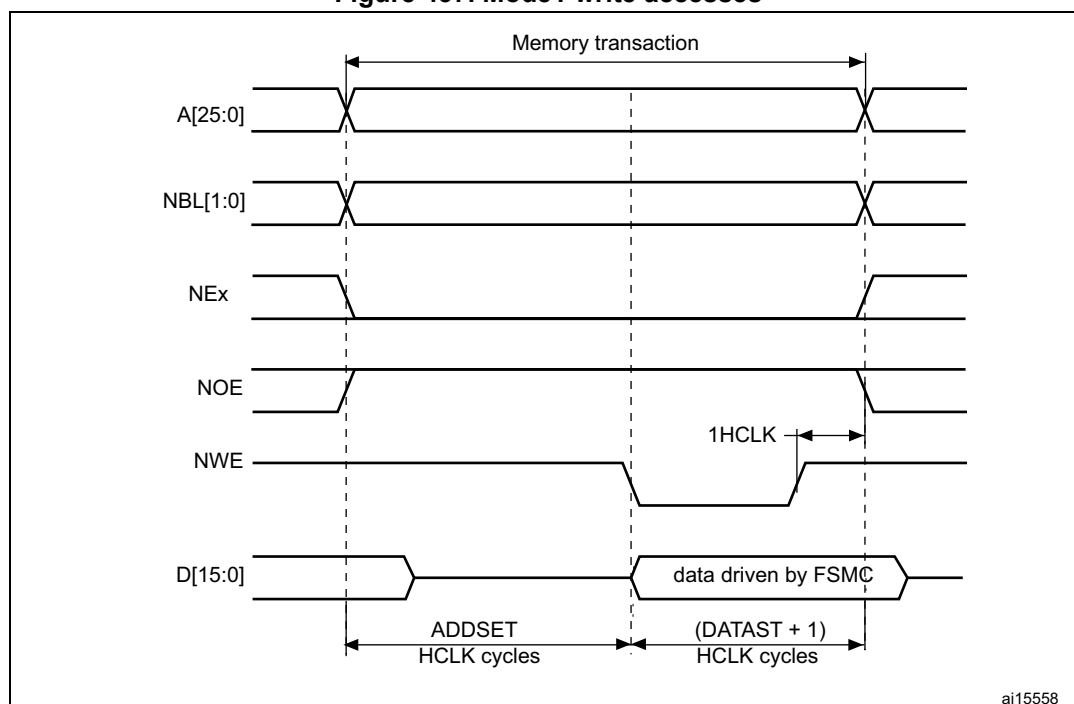
The next figures show the read and write transactions for the supported modes followed by the required configuration of FSMC_BCRx, and FSMC_BTRx/FSMC_BWTRx registers.

Figure 436. Mode1 read accesses



1. NBL[1:0] are driven low during read access.

Figure 437. Mode1 write accesses



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The one HCLK cycle at the end of the write transaction helps guarantee the address and data hold time after the NWE rising edge. Due to the presence of this one HCLK cycle, the DATAST value must be greater than zero ($\text{DATAST} > 0$).

Table 226. FSMC_BCRx bit fields

Bit number	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	0x0 (no effect on asynchronous mode)
18:16	CPSIZE	0x0 (no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise keep at 0.
14	EXTMOD	0x0
13	WAITEN	0x0 (no effect on asynchronous mode)
12	WREN	As needed
11	WAITCFG	Don't care
10	WRAPMOD	0x0
9	WAITPOL	Meaningful only if bit 15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Don't care
5-4	MWID	As needed
3-2	MTYP[0:1]	As needed, exclude 0x2 (NOR flash)

Table 226. FSMC_BCRx bit fields (continued)

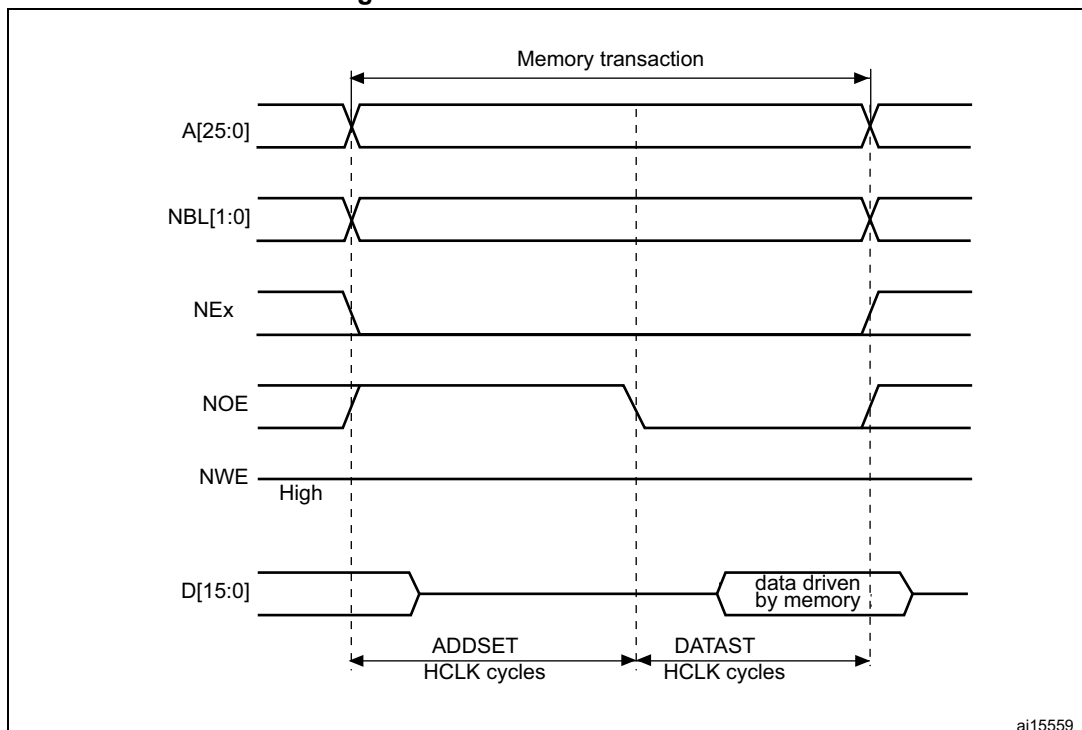
Bit number	Bit name	Value to set
1	MUXE	0x0
0	MBKEN	0x1

Table 227. FSMC_BTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	Don't care
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST+1 HCLK cycles for write accesses, DATAST HCLK cycles for read accesses).
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles). Minimum value for ADDSET is 0.

Mode A - SRAM/PSRAM (CRAM) OE toggling

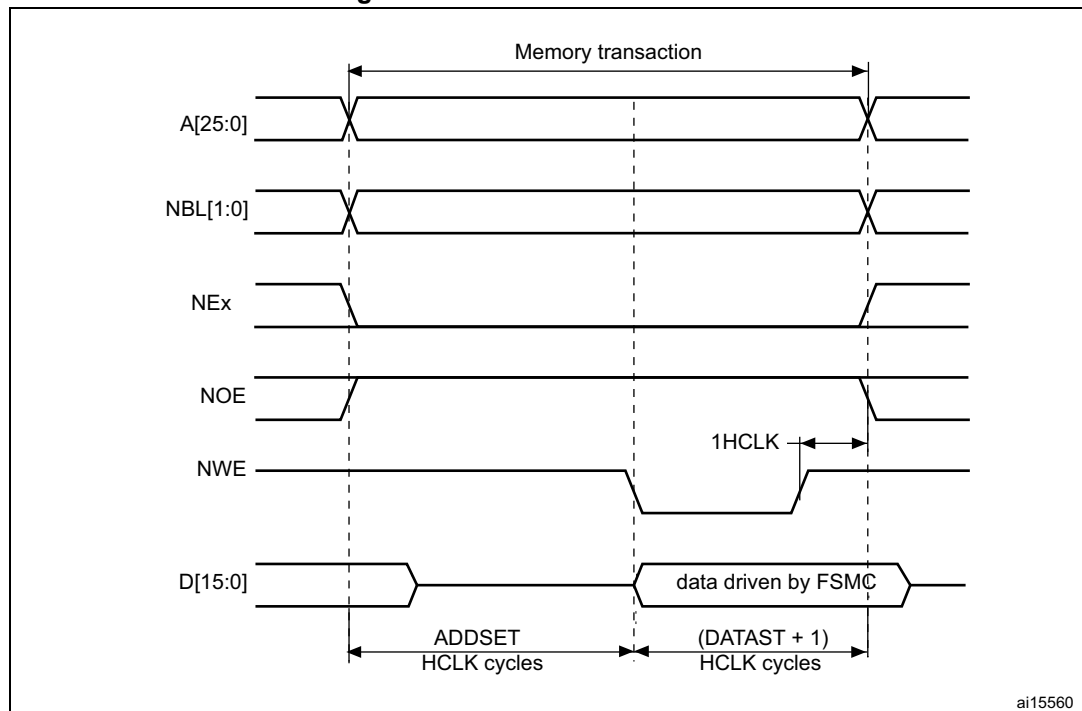
Figure 438. ModeA read accesses



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1. NBL[1:0] are driven low during read access.

Figure 439. ModeA write accesses



The differences compared with mode1 are the toggling of NOE and the independent read and write timings.

Table 228. FSMC_BCRx bit fields

Bit number	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	0x0 (no effect on asynchronous mode)
18:16	CPSIZE	0x0 (no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise keep at 0.
14	EXTMOD	0x1
13	WAITEN	0x0 (no effect on asynchronous mode)
12	WREN	As needed
11	WAITCFG	Don't care
10	WRAPMOD	0x0
9	WAITPOL	Meaningful only if bit 15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Don't care
5-4	MWID	As needed
3-2	MTYP[0:1]	As needed, exclude 0x2 (NOR flash)

Table 228. FSMC_BCRx bit fields (continued)

Bit number	Bit name	Value to set
1	MUXEN	0x0
0	MBKEN	0x1

Table 229. FSMC_BTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x0
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST HCLK cycles) for read accesses.
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0.

Table 230. FSMC_BWTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x0
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST+1 HCLK cycles for write accesses,
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for write accesses. Minimum value for ADDSET is 0.

Mode 2/B - NOR flash

Figure 440. Mode2 and mode B read accesses

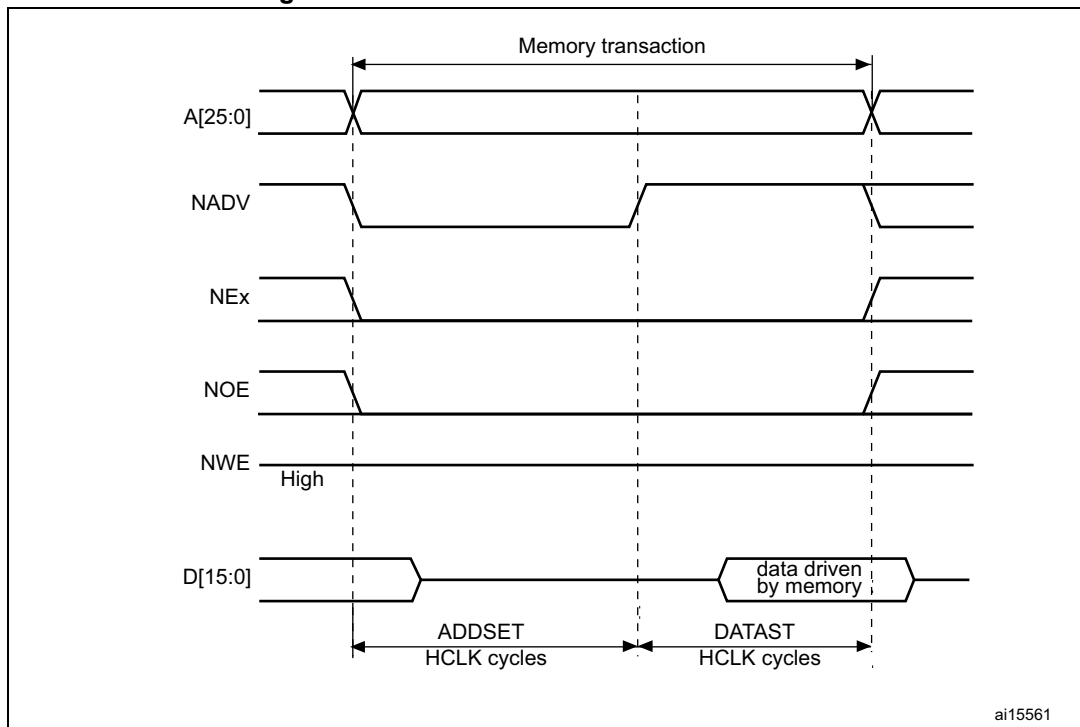


Figure 441. Mode2 write accesses

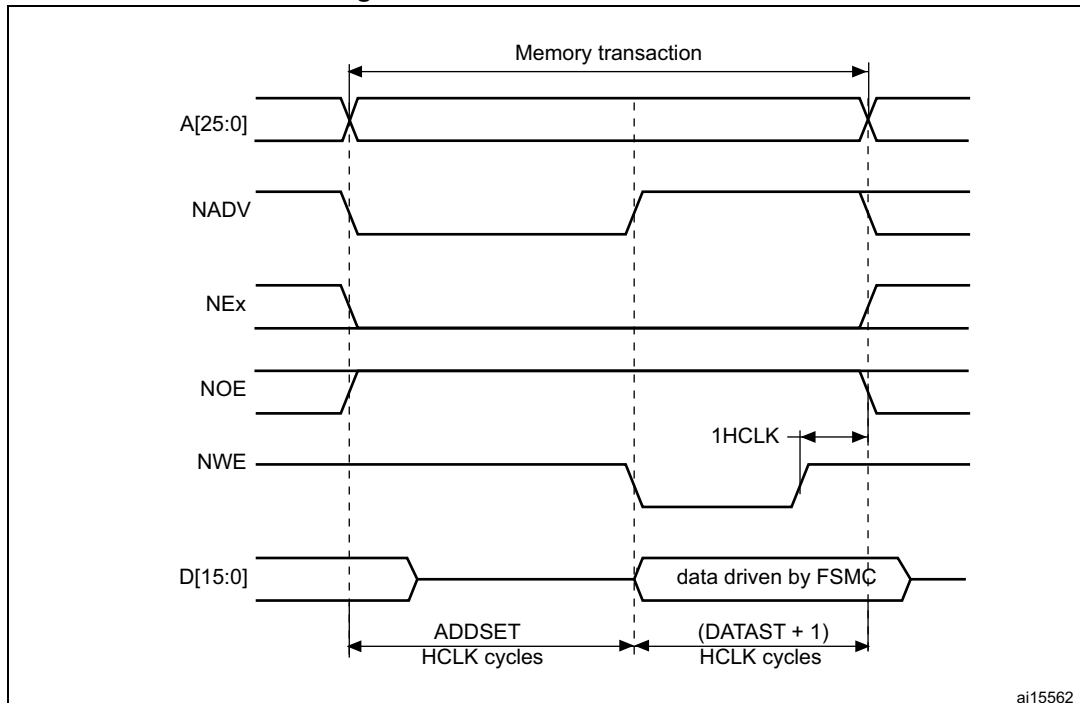
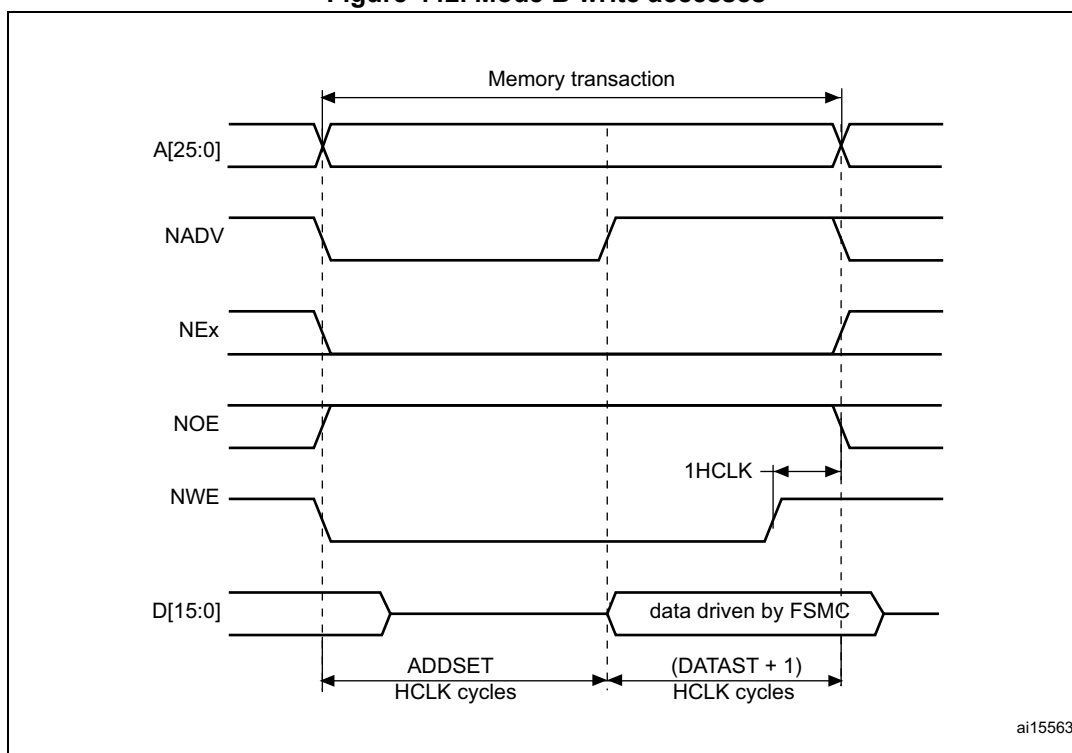


Figure 442. Mode B write accesses



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The differences with mode1 are the toggling of NWE and the independent read and write timings when extended mode is set (Mode B).

Table 231. FSMC_BCRx bit fields

Bit number	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	0x0 (no effect on asynchronous mode)
18:16	Reserved	0x0 (no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise keep at 0.
14	EXTMOD	0x1 for mode B, 0x0 for mode 2
13	WAITEN	0x0 (no effect on asynchronous mode)
12	WREN	As needed
11	WAITCFG	Don't care
10	WRAPMOD	0x0
9	WAITPOL	Meaningful only if bit 15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
5-4	MWID	As needed

Table 231. FSMC_BCRx bit fields (continued)

Bit number	Bit name	Value to set
3-2	MTYP[0:1]	0x2 (NOR flash memory)
1	MUXEN	0x0
0	MBKEN	0x1

Table 232. FSMC_BTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x1
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST HCLK cycles) for read accesses.
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0.

Table 233. FSMC_BWTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x1
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST+1 HCLK cycles for write accesses,
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for write accesses. Minimum value for ADDSET is 0.

Note: The FSMC_BWTRx register is valid only if extended mode is set (mode B), otherwise all its content is don't care.

Mode C - NOR flash - OE toggling

Figure 443. Mode C read accesses

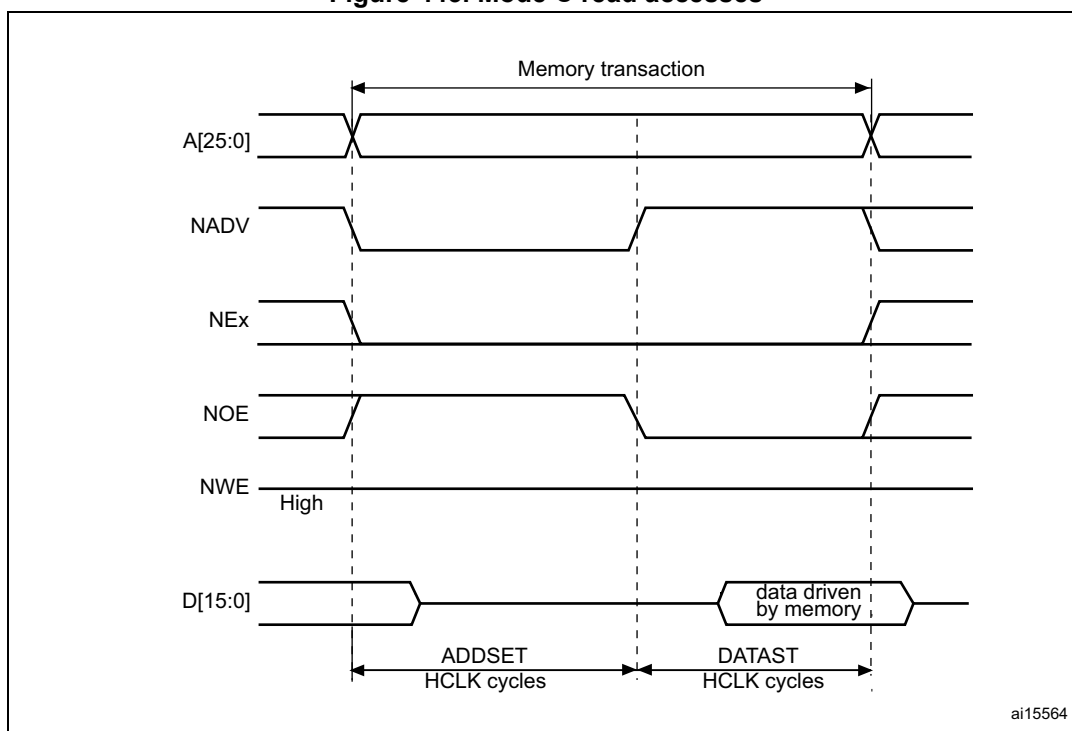
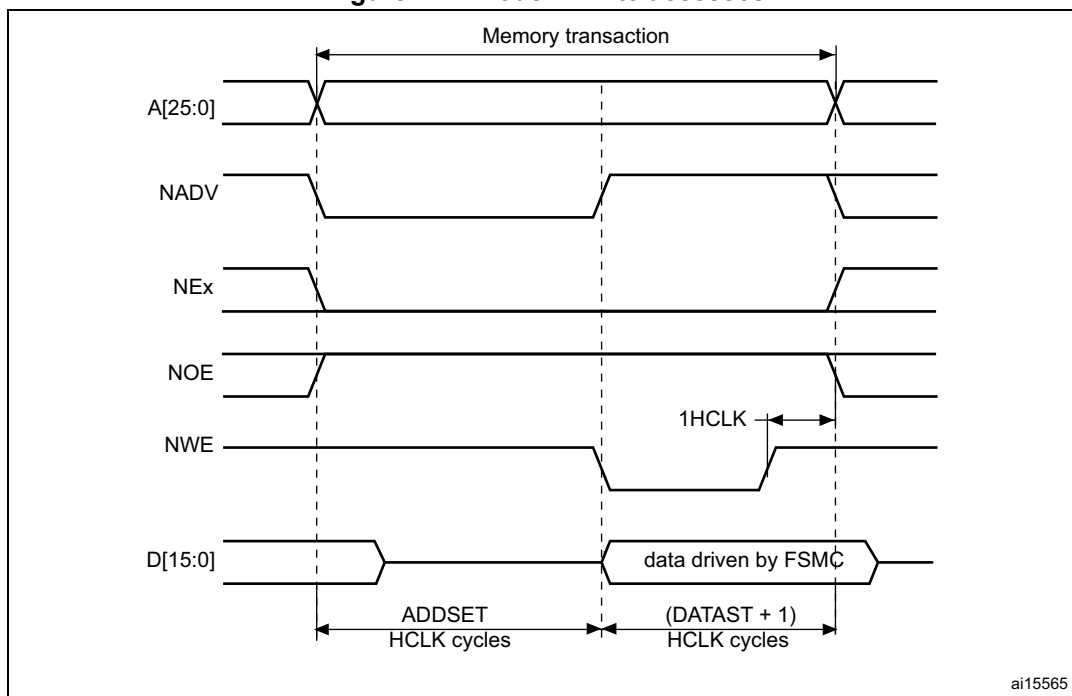


Figure 444. Mode C write accesses



The differences compared with mode1 are the toggling of NOE and the independent read and write timings.

Table 234. FSMC_BCRx bit fields

Bit No.	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	0x0 (no effect on asynchronous mode)
18:16	CPSIZE	0x0 (no effect on asynchronous mode)
15	ASYNCAWAIT	Set to 1 if the memory supports this feature. Otherwise keep at 0.
14	EXTMOD	0x1
13	WAITEN	0x0 (no effect on asynchronous mode)
12	WREN	As needed
11	WAITCFG	Don't care
10	WRAPMOD	0x0
9	WAITPOL	Meaningful only if bit 15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
5-4	MWID	As needed
3-2	MTYP[0:1]	0x2 (NOR flash memory)
1	MUXEN	0x0
0	MBKEN	0x1

Table 235. FSMC_BTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x2
27-24	DATLAT	0x0
23-20	CLKDIV	0x0
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST HCLK cycles) for read accesses.
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0.

Table 236. FSMC_BWTRx bit fields

Bit number	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x2
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST+1 HCLK cycles for write accesses,
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for write accesses. Minimum value for ADDSET is 0.

Mode D - asynchronous access with extended address

Figure 445. Mode D read accesses

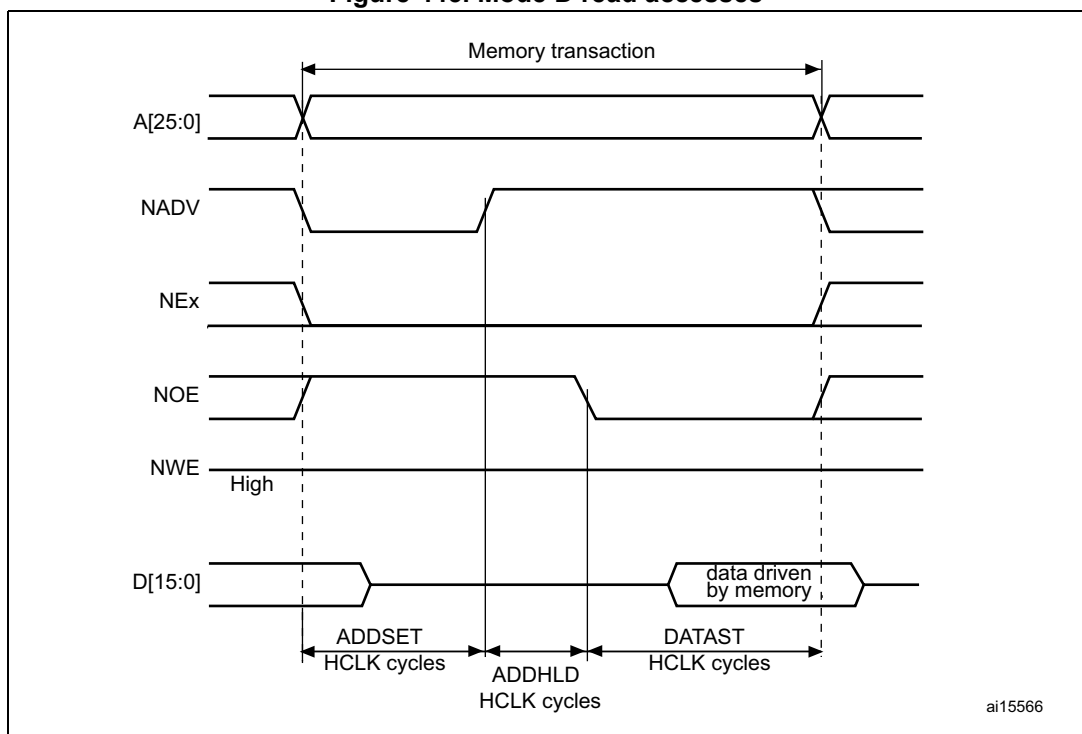
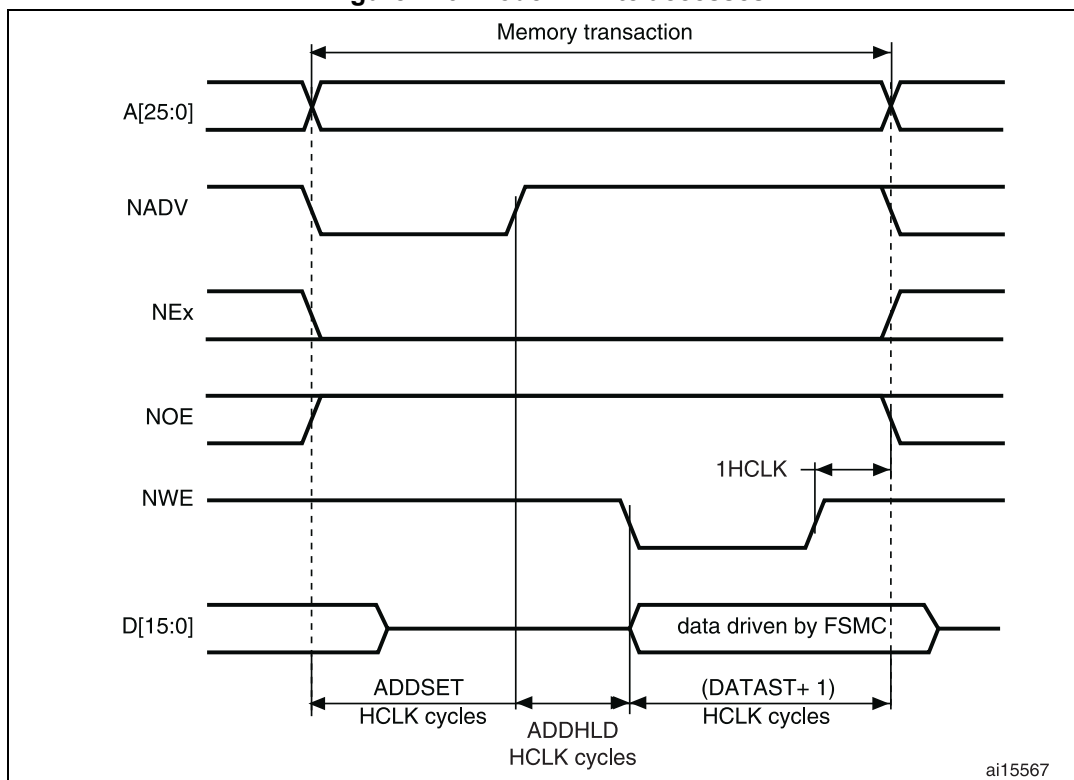


Figure 446. Mode D write accesses



The differences with mode1 are the toggling of NOE that goes on toggling after NADV changes and the independent read and write timings.

Table 237. FSMC_BCRx bit fields

Bit No.	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	0x0 (no effect on asynchronous mode)
18:16	CPSIZE	0x0 (no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise keep at 0.
14	EXTMOD	0x1
13	WAITEN	0x0 (no effect on asynchronous mode)
12	WREN	As needed
11	WAITCFG	Don't care
10	WRAPMOD	0x0
9	WAITPOL	Meaningful only if bit 15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Set according to memory support
5-4	MWID	As needed

Table 237. FSMC_BCRx bit fields (continued)

Bit No.	Bit name	Value to set
3-2	MTYP[0:1]	As needed
1	MUXEN	0x0
0	MBKEN	0x1

Table 238. FSMC_BTRx bit fields

Bit No.	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x3
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST HCLK cycles) for read accesses.
7-4	ADDHLD	Duration of the middle phase of the read access (ADDHLD HCLK cycles)
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0.

Table 239. FSMC_BWTRx bit fields

Bit No.	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x3
27-24	DATLAT	0x0
23-20	CLKDIV	0x0
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST+1 HCLK cycles) for write accesses
7-4	ADDHLD	Duration of the middle phase of the write access (ADDHLD HCLK cycles)
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET+1 HCLK cycles) for write accesses. Minimum value for ADDSET is 0.

Muxed mode - multiplexed asynchronous access to NOR flash memory

Figure 447. Multiplexed read accesses

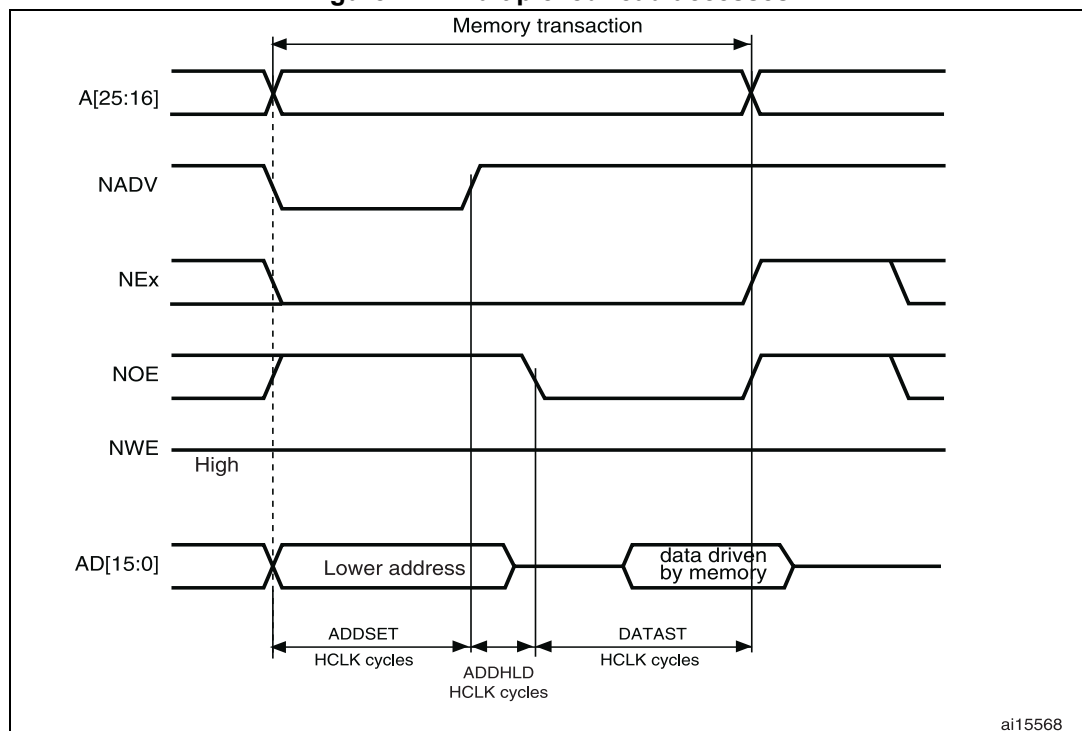
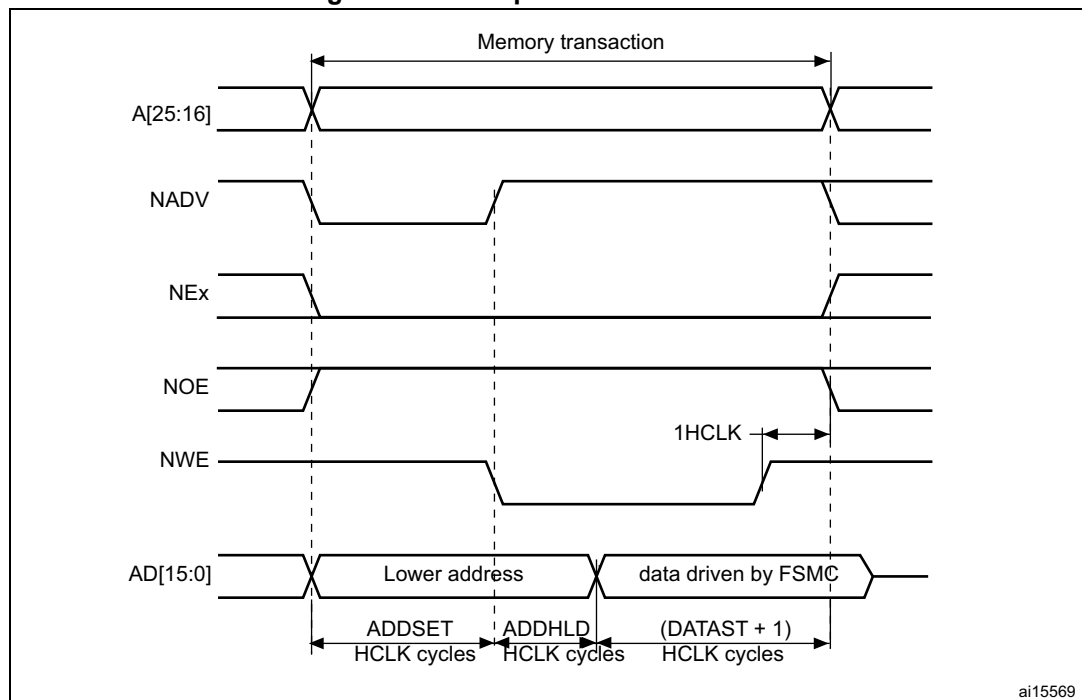


Figure 448. Multiplexed write accesses



The difference with mode D is the drive of the lower address byte(s) on the databus.

Table 240. FSMC_BCRx bit fields

Bit No.	Bit name	Value to set
31-21	Reserved	0x000
19	CBURSTRW	0x0 (no effect on asynchronous mode)
18:16	CPSIZE	0x0 (no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise keep at 0.
14	EXTMOD	0x0
13	WAITEN	0x0 (no effect on asynchronous mode)
12	WREN	As needed
11	WAITCFG	Don't care
10	WRAPMOD	0x0
9	WAITPOL	Meaningful only if bit 15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
5-4	MWID	As needed
3-2	MTYP[0:1]	0x2 (NOR flash memory)
1	MUXEN	0x1
0	MBKEN	0x1

Table 241. FSMC_BTRx bit fields

Bit No.	Bit name	Value to set
31:30	Reserved	0x0
29-28	ACCMOD	0x0
27-24	DATLAT	Don't care
23-20	CLKDIV	Don't care
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Duration of the second access phase (DATAST HCLK cycles for read accesses and DATAST+1 HCLK cycles for write accesses).
7-4	ADDHLD	Duration of the middle phase of the access (ADDHLD HCLK cycles).
3-0	ADDSET[3:0]	Duration of the first access phase (ADDSET HCLK cycles). Minimum value for ADDSET is 1.

WAIT management in asynchronous accesses

If the asynchronous memory asserts a WAIT signal to indicate that it is not yet ready to accept or to provide data, the ASYNCWAIT bit has to be set in FSMC_BCRx register.

If the WAIT signal is active (high or low depending on the WAITPOL bit), the second access phase (Data setup phase) programmed by the DATAST bits, is extended until WAIT becomes inactive. Unlike the data setup phase, the first access phases (Address setup and Address hold phases), programmed by the ADDSET[3:0] and ADDHLD bits, are not WAIT sensitive and so they are not prolonged.

The data setup phase (DATAST in the FSMC_BTRx register) must be programmed so that WAIT can be detected 4 HCLK cycles before the end of memory transaction. The following cases must be considered:

1. DATAST in FSMC_BTRx register) Memory asserts the WAIT signal aligned to NOE/NWE which toggles:

$$\text{DATAST} \geq (4 \times \text{HCLK}) + \text{max_wait_assertion_time}$$

2. Memory asserts the WAIT signal aligned to NEx (or NOE/NWE not toggling):
if

$$\text{max_wait_assertion_time} > \text{address_phase} + \text{hold_phase}$$

then

$$\text{DATAST} \geq (4 \times \text{HCLK}) + (\text{max_wait_assertion_time} - \text{address_phase} - \text{hold_phase})$$

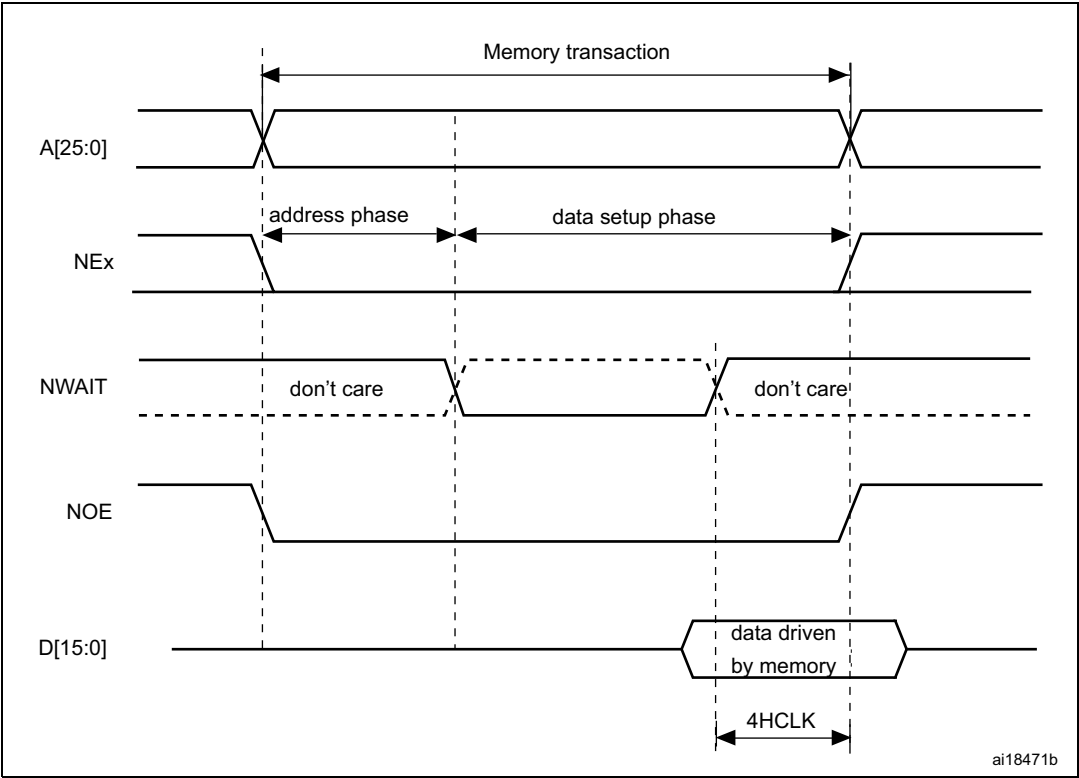
otherwise

$$\text{DATAST} \geq 4 \times \text{HCLK}$$

where max_wait_assertion_time is the maximum time taken by the memory to assert the WAIT signal once NEx/NOE/NWE is low.

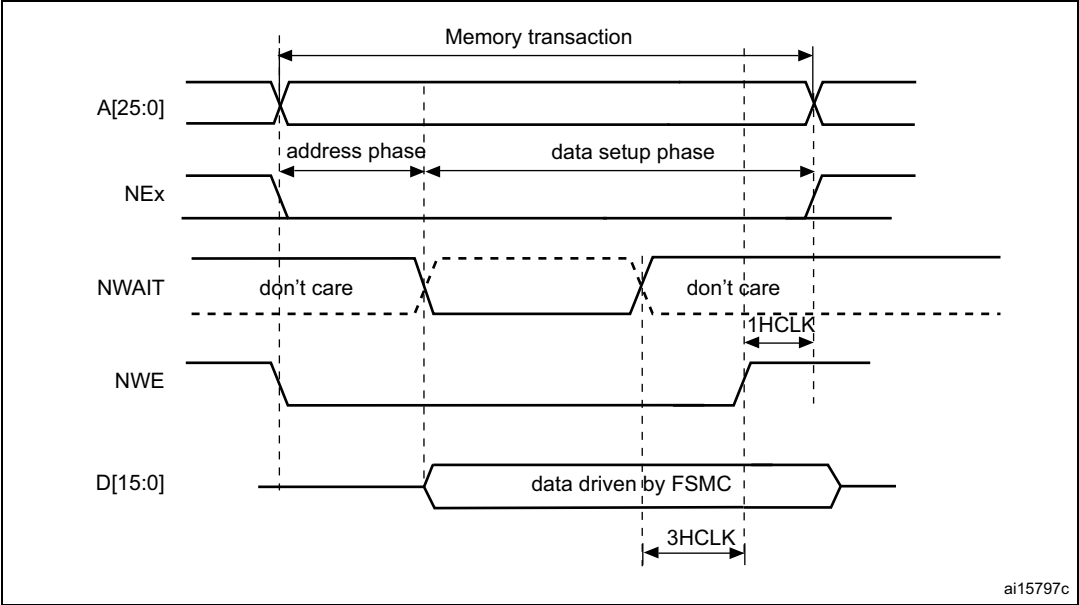
[Figure 449](#) and [Figure 450](#) show the number of HCLK clock cycles that are added to the memory access after WAIT is released by the asynchronous memory (independently of the above cases).

Figure 449. Asynchronous wait during a read access



1. NWAIT polarity depends on WAITPOL bit setting in FSMC_BCRx register.

Figure 450. Asynchronous wait during a write access



1. NWAIT polarity depends on WAITPOL bit setting in FSMC_BCRx register.

36.5.5 Synchronous transactions

The memory clock, CLK, is a submultiple of HCLK according to the value of parameter CLKDIV.

NOR flash memories specify a minimum time from NADV assertion to CLK high. To meet this constraint, the FSMC does not issue the clock to the memory during the first internal clock cycle of the synchronous access (before NADV assertion). This guarantees that the rising edge of the memory clock occurs *in the middle* of the NADV low pulse.

Data latency versus NOR flash latency

The data latency is the number of cycles to wait before sampling the data. The DATLAT value must be consistent with the latency value specified in the NOR flash configuration register. The FSMC does not include the clock cycle when NADV is low in the data latency count.

Caution: Some NOR flash memories include the NADV Low cycle in the data latency count, so the exact relation between the latency and the FMSC DATLAT parameter can be either of:

- NOR flash latency = (DATLAT + 2) CLK clock cycles
- NOR flash latency = (DATLAT + 3) CLK clock cycles

Some recent memories assert NWAIT during the latency phase. In such cases DATLAT can be set to its minimum value. As a result, the FSMC samples the data and waits long enough to evaluate if the data are valid. Thus the FSMC detects when the memory exits latency and real data are taken.

Other memories do not assert NWAIT during latency. In this case the latency must be set correctly for both the FSMC and the memory, otherwise invalid data are mistaken for good data, or valid data are lost in the initial phase of the memory access.

Single-burst transfer

When the selected bank is configured in burst mode for synchronous accesses, if for example an AHB single-burst transaction is requested on 16-bit memories, the FSMC performs a burst transaction of length 1 (if the AHB transfer is 16-bit), or length 2 (if the AHB transfer is 32-bit) and de-assert the chip select signal when the last data is strobed.

Clearly, such a transfer is not the most efficient in terms of cycles (compared to an asynchronous read). Nevertheless, a random asynchronous access would first require to re-program the memory access mode, which would altogether last longer.

Cross boundary page for Cellular RAM 1.5

Cellular RAM 1.5 does not allow burst access to cross the page boundary. The FSMC controller allows to split automatically the burst access when the memory page size is reached by configuring the CPSIZE bits in the FSMC_BCR1 register following the memory page size.

Wait management

For synchronous NOR flash memories, NWAIT is evaluated after the programmed latency period, (DATLAT+2) CLK clock cycles.

If NWAIT is sensed active (low level when WAITPOL = 0, high level when WAITPOL = 1), wait states are inserted until NWAIT is sensed inactive (high level when WAITPOL = 0, low level when WAITPOL = 1).

When NWAIT is inactive, the data is considered valid either immediately (bit WAITCFG = 1) or on the next clock edge (bit WAITCFG = 0).

During wait-state insertion via the NWAIT signal, the controller continues to send clock pulses to the memory, keeping the chip select and output enable signals valid, and does not consider the data valid.

There are two timing configurations for the NOR flash NWAIT signal in burst mode:

- Flash memory asserts the NWAIT signal one data cycle before the wait state (default after reset)
- Flash memory asserts the NWAIT signal during the wait state

These two NOR flash wait state configurations are supported by the FSMC, individually for each chip select, thanks to the WAITCFG bit in the FSMC_BCRx registers (x = 0..3).

Figure 451. Wait configurations

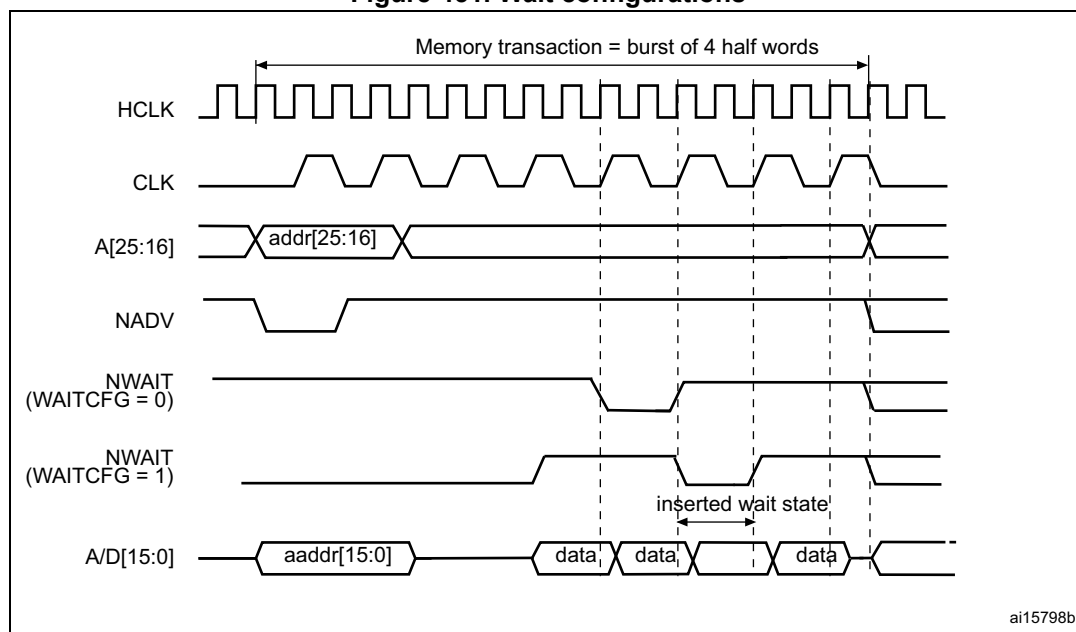
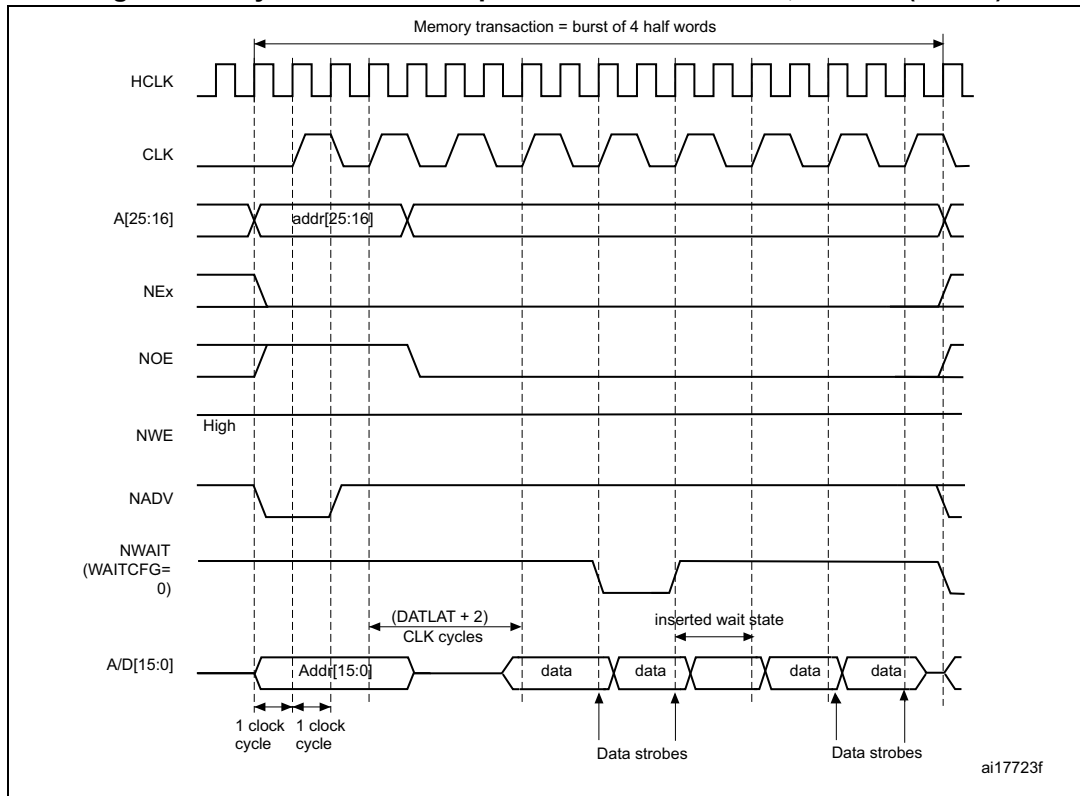


Figure 452. Synchronous multiplexed read mode - NOR, PSRAM (CRAM)

1. Byte lane outputs BL are not shown; for NOR access, they are held high, and, for PSRAM (CRAM) access, they are held low.
2. NWAIT polarity is set to 0.

Table 242. FSMC_BCRx bit fields

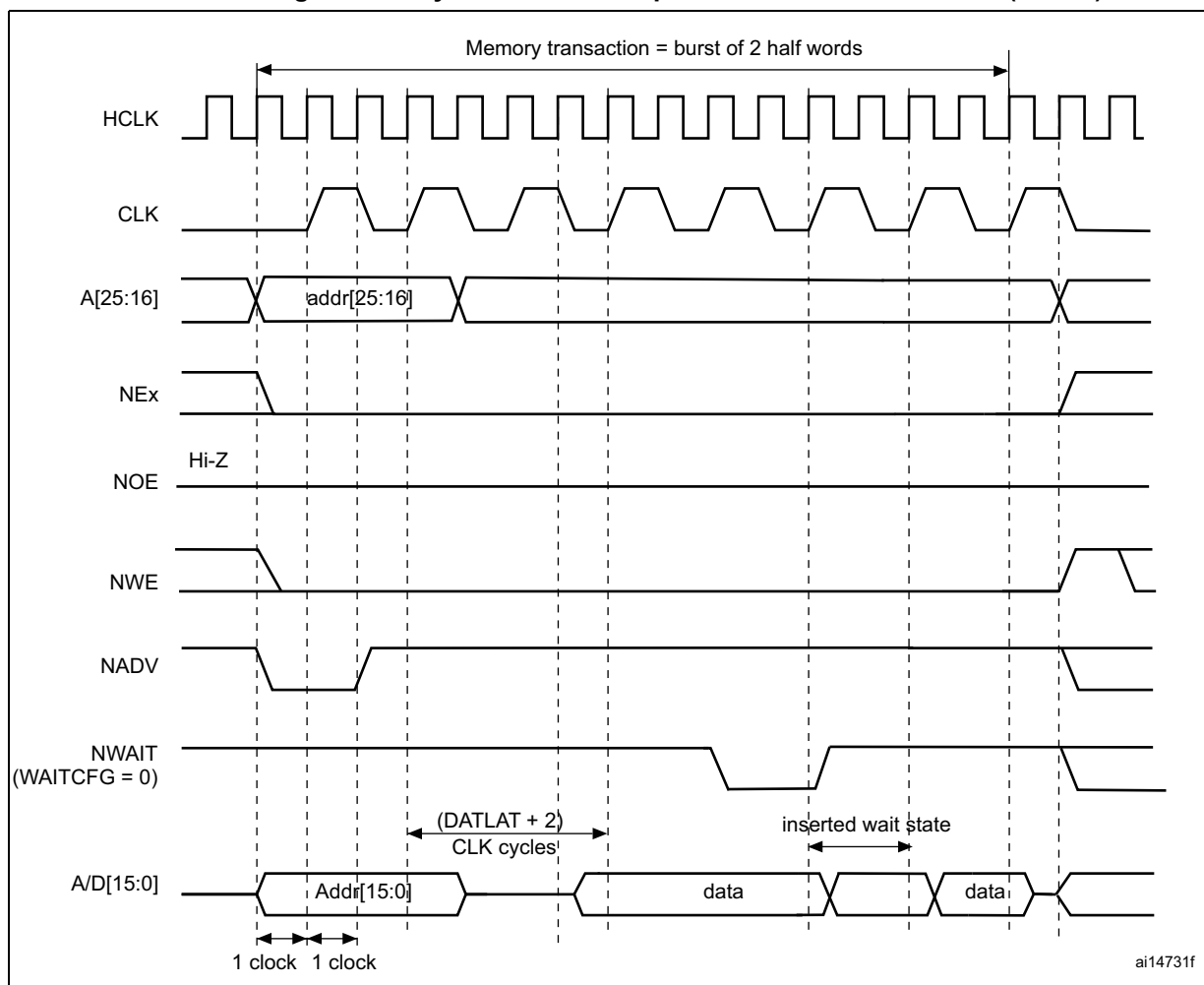
Bit No.	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	No effect on synchronous read
18-16	CPSIZE	As needed (0x1 for CRAM 1.5)
15	ASCYCWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	Set to 1 if the memory supports this feature, otherwise keep at 0.
12	WREN	no effect on synchronous read
11	WAITCFG	to be set according to memory
10	WRAPMOD	0x0
9	WAITPOL	to be set according to memory
8	BURSTEN	0x1
7	Reserved	0x1
6	FACCEN	Set according to memory support (NOR flash memory)
5-4	MWID	As needed

Table 242. FSMC_BCRx bit fields (continued)

Bit No.	Bit name	Value to set
3-2	MTYP[0:1]	0x1 or 0x2
1	MUXEN	As needed
0	MBKEN	0x1

Table 243. FSMC_BTRx bit fields

Bit No.	Bit name	Value to set
31:30	Reserved	0x0
29:28	ACCMOD	0x0
27-24	DATLAT	Data latency
23-20	CLKDIV	0x0 to get CLK = HCLK (not supported) 0x1 to get CLK = 2 × HCLK ..
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Don't care
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Don't care

Figure 453. Synchronous multiplexed write mode - PSRAM (CRAM)

1. Memory must issue NWAIT signal one cycle in advance, accordingly WAITCFG must be programmed to 0.
2. NWAIT polarity is set to 0.
3. Byte Lane (NBL) outputs are not shown, they are held low while NEx is active.

Table 244. FSMC_BCRx bit fields

Bit No.	Bit name	Value to set
31-20	Reserved	0x000
19	CBURSTRW	0x1
18-16	CPSIZE	As needed (0x1 for CRAM 1.5)
15	ASCYCWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	Set to 1 if the memory supports this feature, otherwise keep at 0.
12	WREN	0x1
11	WAITCFG	0x0
10	WRAPMOD	0x0

Table 244. FSMC_BCRx bit fields (continued)

Bit No.	Bit name	Value to set
9	WAITPOL	to be set according to memory
8	BURSTEN	no effect on synchronous write
7	Reserved	0x1
6	FACCEN	Set according to memory support
5-4	MWID	As needed
3-2	MTYP[0:1]	0x1
1	MUXEN	As needed
0	MBKEN	0x1

Table 245. FSMC_BTRx bit fields

Bit No.	Bit name	Value to set
31:30	Reserved	0x0
29:28	ACCMOD	0x0
27-24	DATLAT	Data latency
23-20	CLKDIV	0x0 to get CLK = HCLK (not supported) 0x1 to get CLK = 2 × HCLK
19-16	BUSTURN	Time between NEx high to NEx low (BUSTURN HCLK)
15-8	DATAST	Don't care
7-4	ADDHLD	Don't care
3-0	ADDSET[3:0]	Don't care

36.5.6 NOR/PSRAM control registers

The NOR/PSRAM control registers have to be accessed by words (32 bits).

SRAM/NOR-flash chip-select control registers 1..4 (FSMC_BCR1..4)

Address offset: $0xA000\ 0000 + 8 * (x - 1)$, $x = 1...4$

Reset value: 0x0000 30DB for Bank1 and 0x0000 30D2 for Bank 2 to 4

This register contains the control information of each memory bank, used for SRAMs, PSRAM and NOR flash memories.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CBURSTRW	CPSIZE[2:0]			ASCYCWAIT	EXTMOD	WAITEN	WREN	WAITCFG	WRAPMOD	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]		MTYP[1:0]		MUXEN	MBKEN
												RW				RW	RW	RW	RW	RW	RW	RW	RW			RW	RW	RW	RW	RW	

Bits 31: 20 Reserved, must be kept at reset value.

Bit 19 **CBURSTRW**: Write burst enable.

For Cellular RAM (PSRAM) memories, this bit enables the synchronous burst protocol during write operations. The enable bit for synchronous read accesses is the BURSTEN bit in the FSMC_BCRx register.

0: Write operations are always performed in asynchronous mode

1: Write operations are performed in synchronous mode.

Bits 18: 16 **CPSIZE[2:0]**: CRAM page size.

These are used for Cellular RAM 1.5 which does not allow burst access to cross the address boundaries between pages. When these bits are configured, the FSMC controller splits automatically the burst access when the memory page size is reached (refer to memory datasheet for page size).

000: No burst split when crossing page boundary (default after reset)

001: 128 bytes

010: 256 bytes

011: 512 bytes

100: 1024 bytes

Others: reserved.

Bit 15 **ASYNCAWAIT**: Wait signal during asynchronous transfers

This bit enables/disables the FSMC to use the wait signal even during an asynchronous protocol.

0: NWAIT signal is not taken into account when running an asynchronous protocol (default after reset)

1: NWAIT signal is taken into account when running an asynchronous protocol

Bit 14 EXTMOD: Extended mode enable.

This bit enables the FSMC to program the write timings for non-multiplexed asynchronous accesses inside the FSMC_BWTR register, thus resulting in different timings for read and write operations.

0: values inside FSMC_BWTR register are not taken into account (default after reset)

1: values inside FSMC_BWTR register are taken into account

Note: When the extended mode is disabled, the FSMC can operate in Mode1 or Mode2 as follows:

- *Mode 1 is the default mode when the SRAM/PSRAM memory type is selected (MTYP [0:1]=0x0 or 0x01)*
- *Mode 2 is the default mode when the NOR memory type is selected (MTYP [0:1]= 0x10).*

Bit 13 WAITEN: Wait enable bit.

This bit enables/disables wait-state insertion via the NWAIT signal when accessing the flash memory in synchronous mode.

0: NWAIT signal is disabled (its level not taken into account, no wait state inserted after the programmed flash latency period)

1: NWAIT signal is enabled (its level is taken into account after the programmed flash latency period to insert wait states if asserted) (default after reset)

Bit 12 WREN: Write enable bit.

This bit indicates whether write operations are enabled/disabled in the bank by the FSMC:

0: Write operations are disabled in the bank by the FSMC, an AHB error is reported,

1: Write operations are enabled for the bank by the FSMC (default after reset).

Bit 11 WAITCFG: Wait timing configuration.

The NWAIT signal indicates whether the data from the memory are valid or if a wait state must be inserted when accessing the flash memory in synchronous mode. This configuration bit determines if NWAIT is asserted by the memory one clock cycle before the wait state or during the wait state:

0: NWAIT signal is active one data cycle before wait state (default after reset),

1: NWAIT signal is active during wait state (not used for PRAM).

Bit 10 WRAPMOD: Wrapped burst mode support.

Defines whether the controller splits or not an AHB burst wrap access into two linear accesses. Valid only when accessing memories in burst mode

0: Direct wrapped burst is not enabled (default after reset),

1: Direct wrapped burst is enabled.

Note: This bit has no effect as the CPU and DMA cannot generate wrapping burst transfers.

Bit 9 WAITPOL: Wait signal polarity bit.

Defines the polarity of the wait signal from memory. Valid only when accessing the memory in burst mode:

0: NWAIT active low (default after reset),

1: NWAIT active high.

Bit 8 BURSTEN: Burst enable bit.

This bit enables/disables synchronous accesses during read operations. It is valid only for synchronous memories operating in burst mode:

0: Burst mode disabled (default after reset). Read accesses are performed in asynchronous mode.

1: Burst mode enable. Read accesses are performed in synchronous mode.

Bit 7 Reserved, must be kept at reset value.

Bit 6 **FACCEN**: Flash access enable

Enables NOR flash memory access operations.

0: Corresponding NOR flash memory access is disabled

1: Corresponding NOR flash memory access is enabled (default after reset)

Bits 5:4 **MWID[1:0]**: Memory databus width.

Defines the external memory device width, valid for all type of memories.

00: 8 bits,

01: 16 bits (default after reset),

10: reserved, do not use,

11: reserved, do not use.

Bits 3:2 **MTYP[1:0]**: Memory type.

Defines the type of external memory attached to the corresponding memory bank:

00: SRAM (default after reset for Bank 2...4)

01: PSRAM (CRAM)

10: NOR flash/OneNAND flash (default after reset for Bank 1)

11: reserved

Bit 1 **MUXEN**: Address/data multiplexing enable bit.

When this bit is set, the address and data values are multiplexed on the databus, valid only with NOR and PSRAM memories:

0: Address/Data nonmultiplexed

1: Address/Data multiplexed on databus (default after reset)

Bit 0 **MBKEN**: Memory bank enable bit.

Enables the memory bank. After reset Bank1 is enabled, all others are disabled.

Accessing a disabled bank causes an ERROR on AHB bus.

0: Corresponding memory bank is disabled

1: Corresponding memory bank is enabled

SRAM/NOR-flash chip-select timing registers 1..4 (FSMC_BTR1..4)

Address offset: $0xA000\ 0000 + 0x04 + 8 * (x - 1)$, $x = 1..4$

Reset value: 0x0FFF FFFF

FSMC_BTRx bits are written by software to add a delay at the end of a read /write transaction. This delay allows matching the minimum time between consecutive transactions (t_{EHEL} from NEx high to FSMC_NEx low) and the maximum time required by the memory to free the data bus after a read access (t_{EHQZ}).

This register contains the control information of each memory bank, used for SRAMs, PSRAM and NOR flash memories. If the EXTMOD bit is set in the FSMC_BCRx register, then this register is partitioned for write and read access, that is, 2 registers are available: one to configure read accesses (this register) and one to configure write accesses (FSMC_BWTRx registers).

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		ACCMOD[1:0]		DATLAT[3:0]				CLKDIV[3:0]				BUSTURN[3:0]				DATAST[7:0]							ADDHLD[3:0]				ADDSET[3:0]				
		rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:28 **ACCMOD[1:0]**: Access mode

Specifies the asynchronous access modes as shown in the timing diagrams. These bits are taken into account only when the EXTMOD bit in the FSMC_BCRx register is 1.

- 00: access mode A
- 01: access mode B
- 10: access mode C
- 11: access mode D

Bits 27:24 **DATLAT[3:0]**: Data latency for synchronous memory (see note below bit description table)

For synchronous accesses with read/write burst mode enabled (BURSTEN / CBURSTRW bits set), this field defines the number of memory clock cycles (+2) to issue to the memory before reading/writing the first data. This timing parameter is not expressed in HCLK periods, but in FSMC_CLK periods. For asynchronous accesses, this value is don't care.

- 0000: Data latency of 2 CLK clock cycles for first burst access
- 1111: Data latency of 17 CLK clock cycles for first burst access (default value after reset)

Bits 23:20 **CLKDIV[3:0]**: Clock divide ratio (for FSMC_CLK signal)

Defines the period of FSMC_CLK clock output signal, expressed in number of HCLK cycles:

- 0000: Reserved
- 0001: FSMC_CLK period = 2 × HCLK periods
- 0010: FSMC_CLK period = 3 × HCLK periods
- 1111: FSMC_CLK period = 16 × HCLK periods (default value after reset)

In asynchronous NOR flash, SRAM or PSRAM accesses, this value is don't care.

Bits 19:16 **BUSTURN[3:0]**: Bus turnaround phase duration

These bits are written by software to add a delay at the end of a write-to-read (and read-to-write) transaction. The programmed bus turnaround delay is inserted between an asynchronous read (muxed or D mode) or a write transaction and any other asynchronous/synchronous read or write to/from a static bank (for a read operation, the bank can be the same or a different one; for a write operation, the bank can be different except in r muxed or D mode).

In some cases, the bus turnaround delay is fixed, whatever the programmed BUSTURN values:

- No bus turnaround delay is inserted between two consecutive asynchronous write transfers to the same static memory bank except in muxed and D mode.
- A bus turnaround delay of 1 FSMC clock cycle is inserted between:
 - Two consecutive asynchronous read transfers to the same static memory bank except for muxed and D modes.
 - An asynchronous read to an asynchronous or synchronous write to any static bank or dynamic bank except for muxed and D modes.
 - An asynchronous (modes 1, 2, A, B or C) read and a read operation from another static bank.
- A bus turnaround delay of 2 FSMC clock cycles is inserted between:
 - Two consecutive synchronous write accesses (in burst or single mode) to the same bank
 - A synchronous write (burst or single) access and an asynchronous write or read transfer to or from static memory bank (the bank can be the same or different in case of a read operation).
 - Two consecutive synchronous read accesses (in burst or single mode) followed by a any synchronous/asynchronous read or write from/to another static memory bank.
- A bus turnaround delay of 3 FSMC clock cycles is inserted between:
 - Two consecutive synchronous write operations (in burst or single mode) to different static banks.
 - A synchronous write access (in burst or single mode) and a synchronous read access from the same or to a different bank.

0000: BUSTURN phase duration = 0 HCLK clock cycle added

...

1111: BUSTURN phase duration = 15 × HCLK clock cycles (default value after reset)

Bits 15:8 **DATAST[7:0]**: Data-phase duration

These bits are written by software to define the duration of the data phase (refer to [Figure 436](#) to [Figure 448](#)), used in asynchronous accesses:

0000 0000: Reserved

0000 0001: DATAST phase duration = 1 × HCLK clock cycles

0000 0010: DATAST phase duration = 2 × HCLK clock cycles

...

1111 1111: DATAST phase duration = 255 × HCLK clock cycles (default value after reset)

For each memory type and access mode data-phase duration, refer to the respective figure ([Figure 436](#) to [Figure 448](#)).

Example: Mode1, write access, DATAST=1: Data-phase duration= DATAST+1 = 2 HCLK clock cycles.

Note: In synchronous accesses, this value is don't care.

Bits 7:4 **ADDHLD[3:0]**: Address-hold phase duration

These bits are written by software to define the duration of the *address hold* phase (refer to [Figure 445](#) to [Figure 448](#)), used in mode D and multiplexed accesses:

0000: Reserved

0001: ADDHLD phase duration = 1 × HCLK clock cycle

0010: ADDHLD phase duration = 2 × HCLK clock cycle

...

1111: ADDHLD phase duration = 15 × HCLK clock cycles (default value after reset)

For each access mode address-hold phase duration, refer to the respective figure ([Figure 445](#) to [Figure 448](#)).

Note: In synchronous accesses, this value is not used, the address hold phase is always 1 memory clock period duration.

Bits 3:0 **ADDSET[3:0]**: Address setup phase duration

These bits are written by software to define the duration of the *address setup* phase (refer to [Figure 436](#) to [Figure 448](#)), used in SRAMs, ROMs and asynchronous NOR flash and PSRAM accesses:

0000: ADDSET phase duration = 0 × HCLK clock cycle

...

1111: ADDSET phase duration = 15 × HCLK clock cycles (default value after reset)

For each access mode address setup phase duration, refer to the respective figure (refer to [Figure 436](#) to [Figure 448](#)).

Note: In synchronous NOR flash and PSRAM accesses, this value is don't care.

Note: *PSRAMs (CRAMs) have a variable latency due to internal refresh. Therefore these memories issue the NWAIT signal during the whole latency phase to prolong the latency as needed.*

With PSRAMs (CRAMs) the DATLAT field must be set to 0, so that the FSMC exits its latency phase soon and starts sampling NWAIT from memory, then starts to read or write when the memory is ready.

This method can be used also with the latest generation of synchronous flash memories that issue the NWAIT signal, unlike older flash memories (check the datasheet of the specific flash memory being used).

SRAM/NOR-flash write timing registers 1..4 (FSMC_BWTR1..4)

Address offset: $0xA000\ 0000 + 0x104 + 8 * (x - 1)$, $x = 1...4$

Reset value: 0x0FFF FFFF

This register contains the control information of each memory bank, used for SRAMs, PSRAMs and NOR flash memories. This register is active for write asynchronous access only when the EXTMOD bit is set in the FSMC_BCRx register.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	ACCM OD[2:0]		Reserved									BUSTURN[3:0]				DATAST[7:0]								ADDHLD[3:0]				ADDSET[3:0]			
	rW	rW										rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:28 **ACCMOD[2:0]**: Access mode.

Specifies the asynchronous access modes as shown in the next timing diagrams. These bits are taken into account only when the EXTMOD bit in the FSMC_BCRx register is 1.

- 00: access mode A
- 01: access mode B
- 10: access mode C
- 11: access mode D

Bits 27:20 Reserved, must be kept at reset value.

Bits 19:16 **BUSTURN[3:0]**: Bus turnaround phase duration

The programmed bus turnaround delay is inserted between a an asynchronous write transfer and any other asynchronous/synchronous read or write transfer to/from a static bank (for a read operation, the bank can be the same or a different one; for a write operation, the bank can be different except in r muxed or D mode).

In some cases, the bus turnaround delay is fixed, whatever the programmed BUSTURN values:

- No bus turnaround delay is inserted between two consecutive asynchronous write transfers to the same static memory bank except in muxed and D mode.
- A bus turnaround delay of 2 FSMC clock cycles is inserted between:
 - Two consecutive synchronous write accesses (in burst or single mode) to the same bank.
 - A synchronous write transfer (in burst or single mode) and an asynchronous write or read transfer to/from static a memory bank.
- A bus turnaround delay of 3 FSMC clock cycles is inserted between:
 - Two consecutive synchronous write accesses (in burst or single mode) to different static banks.
 - A synchronous write transfer (in burst or single mode) and a synchronous read from the same or from a different bank.

0000: BUSTURN phase duration = 0 HCLK clock cycle added

...

1111: BUSTURN phase duration = 15 HCLK clock cycles added (default value after reset)

Bits 15:8 **DATAST[7:0]**: Data-phase duration.

These bits are written by software to define the duration of the data phase (refer to [Figure 436](#) to [Figure 448](#)), used in asynchronous SRAM, PSRAM and NOR flash memory accesses:

0000 0000: Reserved

0000 0001: DATAST phase duration = 1 × HCLK clock cycles

0000 0010: DATAST phase duration = 2 × HCLK clock cycles

...

1111 1111: DATAST phase duration = 255 × HCLK clock cycles (default value after reset)

Note: In synchronous accesses, this value is don't care.

Bits 7:4 **ADDHLD[3:0]**: Address-hold phase duration.

These bits are written by software to define the duration of the *address hold* phase (refer to [Figure 445](#) to [Figure 448](#)), used in asynchronous multiplexed accesses:

0000: Reserved

0001: ADDHLD phase duration = 1 × HCLK clock cycle

0010: ADDHLD phase duration = 2 × HCLK clock cycle

...

1111: ADDHLD phase duration = 15 × HCLK clock cycles (default value after reset)

Note: In synchronous NOR flash accesses, this value is not used, the address hold phase is always 1 flash clock period duration.

Bits 3:0 **ADDSET[3:0]**: Address setup phase duration.

These bits are written by software to define the duration of the *address setup* phase in HCLK cycles (refer to [Figure 445](#) to [Figure 448](#)), used in asynchronous accessed:

0000: ADDSET phase duration = 0 × HCLK clock cycle

...

1111: ADDSET phase duration = 15 × HCLK clock cycles (default value after reset)

Note: In synchronous NOR flash and PSRAM accesses, this value is don't care.

36.6 NAND flash/PC Card controller

The FSMC generates the appropriate signal timings to drive the following types of device:

- NAND flash
 - 8-bit
 - 16-bit
- 16-bit PC Card compatible devices

The NAND/PC Card controller can control three external banks. Bank 2 and bank 3 support NAND flash devices. Bank 4 supports PC Card devices.

Each bank is configured by means of dedicated registers ([Section 36.6.8](#)). The programmable memory parameters include access timings (shown in [Table 246](#)) and ECC configuration.

Table 246. Programmable NAND/PC Card access parameters

Parameter	Function	Access mode	Unit	Min.	Max.
Memory setup time	Number of clock cycles (HCLK) to set up the address before the command assertion	Read/Write	AHB clock cycle (HCLK)	1	255
Memory wait	Minimum duration (HCLK clock cycles) of the command assertion	Read/Write	AHB clock cycle (HCLK)	2	256
Memory hold	Number of clock cycles (HCLK) to hold the address (and the data in case of a write access) after the command de-assertion	Read/Write	AHB clock cycle (HCLK)	1	254
Memory databus high-Z	Number of clock cycles (HCLK) during which the databus is kept in high-Z state after the start of a write access	Write	AHB clock cycle (HCLK)	0	255

36.6.1 External memory interface signals

The following tables list the signals that are typically used to interface NAND flash and PC Card.

Note: Prefix “N” specifies the associated signal as active low.

8-bit NAND flash

Table 247. 8-bit NAND flash

FSMC signal name	I/O	Function
A[17]	O	NAND flash address latch enable (ALE) signal
A[16]	O	NAND flash command latch enable (CLE) signal
D[7:0]	I/O	8-bit multiplexed, bidirectional address/data bus
NCE[x]	O	Chip select, x = 2, 3
NOE(= NRE)	O	Output enable (memory signal name: read enable, NRE)
NWE	O	Write enable
NWAIT/INT[3:2]	I	NAND flash ready/busy input signal to the FSMC

There is no theoretical capacity limitation as the FSMC can manage as many address cycles as needed.

16-bit NAND flash**Table 248. 16-bit NAND flash**

FSMC signal name	I/O	Function
A[17]	O	NAND flash address latch enable (ALE) signal
A[16]	O	NAND flash command latch enable (CLE) signal
D[15:0]	I/O	16-bit multiplexed, bidirectional address/data bus
NCE[x]	O	Chip select, x = 2, 3
NOE(= NRE)	O	Output enable (memory signal name: read enable, NRE)
NWE	O	Write enable
NWAIT/INT[3:2]	I	NAND flash ready/busy input signal to the FSMC

There is no theoretical capacity limitation as the FSMC can manage as many address cycles as needed.

16-bit PC Card**Table 249. 16-bit PC Card**

FSMC signal name	I/O	Function
A[10:0]	O	Address bus
NIORD	O	Output enable for I/O space
NIOWR	O	Write enable for I/O space
NREG	O	Register signal indicating if access is in Common or Attribute space
D[15:0]	I/O	Bidirectional databus
NCE4_1	O	Chip select 1
NCE4_2	O	Chip select 2 (indicates if access is 16-bit or 8-bit)
NOE	O	Output enable in Common and in Attribute space
NWE	O	Write enable in Common and in Attribute space
NWAIT	I	PC Card wait input signal to the FSMC (memory signal name IORDY)
INTR	I	PC Card interrupt to the FSMC (only for PC Cards that can generate an interrupt)
CD	I	PC Card presence detection. Active high. If an access is performed to the PC Card banks while CD is low, an AHB error is generated. Refer to Section 36.3: AHB interface

36.6.2 NAND flash / PC Card supported memories and transactions

Table 250 below shows the supported devices, access modes and transactions. Transactions not allowed (or not supported) by the NAND flash / PC Card controller appear in gray.

Table 250. Supported memories and transactions

Device	Mode	R/W	AHB data size	Memory data size	Allowed/not allowed	Comments
NAND 8-bit	Asynchronous	R	8	8	Y	-
	Asynchronous	W	8	8	Y	-
	Asynchronous	R	16	8	Y	Split into 2 FSMC accesses
	Asynchronous	W	16	8	Y	Split into 2 FSMC accesses
	Asynchronous	R	32	8	Y	Split into 4 FSMC accesses
	Asynchronous	W	32	8	Y	Split into 4 FSMC accesses
NAND 16-bit	Asynchronous	R	8	16	Y	-
	Asynchronous	W	8	16	N	-
	Asynchronous	R	16	16	Y	-
	Asynchronous	W	16	16	Y	-
	Asynchronous	R	32	16	Y	Split into 2 FSMC accesses
	Asynchronous	W	32	16	Y	Split into 2 FSMC accesses

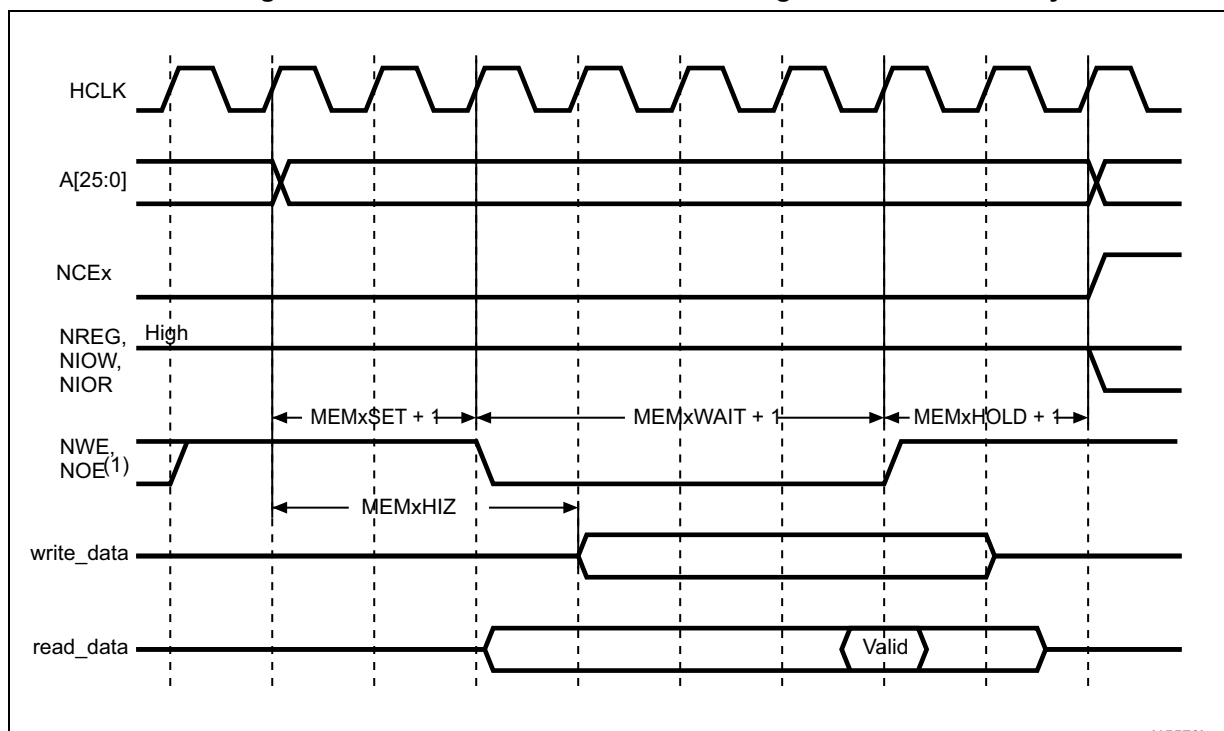
36.6.3 Timing diagrams for NAND and PC Card

Each PC Card/CompactFlash and NAND flash memory bank is managed through a set of registers:

- Control register: FSMC_PCRx
- Interrupt status register: FSMC_SRx
- ECC register: FSMC_ECCRx
- Timing register for Common memory space: FSMC_PMEMx
- Timing register for Attribute memory space: FSMC_PATTx
- Timing register for I/O space: FSMC_PIOx

Each timing configuration register contains three parameters used to define number of HCLK cycles for the three phases of any PC Card/CompactFlash or NAND flash access, plus one parameter that defines the timing for starting driving the databus in the case of a write. [Figure 454](#) shows the timing parameter definitions for common memory accesses, knowing that Attribute and I/O (only for PC Card) memory space access timings are similar.

Figure 454. NAND/PC Card controller timing for common memory access



1. NOE remains high (inactive) during write access. NWE remains high (inactive) during read access.
2. For write accesses, the hold phase delay is $(\text{MEMHOLD}) \times \text{HCLK}$ cycles, while it is $(\text{MEMHOLD} + 2) \times \text{HCLK}$ cycles for read accesses.

36.6.4 NAND flash operations

The command latch enable (CLE) and address latch enable (ALE) signals of the NAND flash device are driven by some address signals of the FSMC controller. This means that to send a command or an address to the NAND flash memory, the CPU has to perform a write to a certain address in its memory space.

A typical page read operation from the NAND flash device is as follows:

1. Program and enable the corresponding memory bank by configuring the FSMC_PCRx and FSMC_PMEMx (and for some devices, FSMC_PATTx, see [Section 36.6.5](#)) registers according to the characteristics of the NAND flash (PWID bits for the databus width of the NAND flash, PTYP = 1, PWAITEN = 0 or 1 as needed, see [Common memory space timing register 2.4 \(FSMC_PMEM2.4\)](#) for timing configuration).
2. The CPU performs a byte write in the common memory space, with data byte equal to one flash command byte (for example 0x00 for Samsung NAND flash devices). The CLE input of the NAND flash is active during the write strobe (low pulse on NWE), thus the written byte is interpreted as a command by the NAND flash. Once the command is latched by the NAND flash device, it does not need to be written for the following page read operations.
3. The CPU can send the start address (STARTAD) for a read operation by writing the required bytes (for example four bytes or three for smaller capacity devices), STARTAD[7:0], STARTAD[15:8], STARTAD[23:16] and finally STARTAD[25:24] for 64 Mb x 8 bit NAND flash) in the common memory or attribute space. The ALE input of the NAND flash device is active during the write strobe (low pulse on NWE), thus the

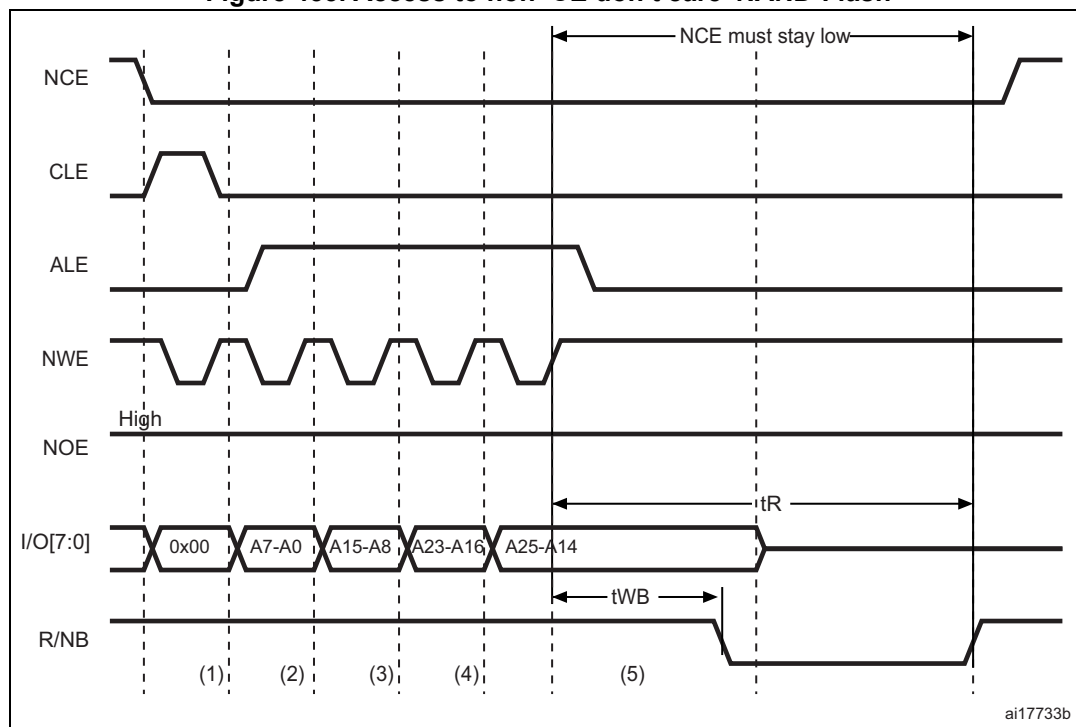
written bytes are interpreted as the start address for read operations. Using the attribute memory space makes it possible to use a different timing configuration of the FSMC, which can be used to implement the prewait functionality needed by some NAND flash memories (see details in [Section 36.6.5](#)).

4. The controller waits for the NAND flash to be ready (R/NB signal high) to become active, before starting a new access (to same or another memory bank). While waiting, the controller maintains the NCE signal active (low).
5. The CPU can then perform byte read operations in the common memory space to read the NAND flash page (data field + Spare field) byte by byte.
6. The next NAND flash page can be read without any CPU command or address write operation, in three different ways:
 - by simply performing the operation described in step 5
 - a new random address can be accessed by restarting the operation at step 3
 - a new command can be sent to the NAND flash device by restarting at step 2

36.6.5 NAND flash prewait functionality

Some NAND flash devices require that, after writing the last part of the address, the controller wait for the R/NB signal to go low as shown in [Figure 455](#).

Figure 455. Access to non 'CE don't care' NAND-Flash



1. CPU wrote byte 0x00 at address 0x7001 0000.
2. CPU wrote byte A7-A0 at address 0x7002 0000.
3. CPU wrote byte A15-A8 at address 0x7002 0000.
4. CPU wrote byte A23-A16 at address 0x7002 0000.
5. CPU wrote byte A25-A24 at address 0x7802 0000: FSMC performs a write access using FSMC_PATT2 timing definition, where $ATTHOLD \geq 7$ (providing that $(7+1) \times HCLK = 112 \text{ ns} > t_{WB} \text{ max}$). This guarantees that NCE remains low until R/NB goes low and high again (only requested for NAND flash memories where NCE is not don't care).

When this functionality is needed, it can be guaranteed by programming the MEMHOLD value to meet the t_{WB} timing. However CPU read accesses to the NAND flash memory has a hold delay of $(MEMHOLD + 2) \times HCLK$ cycles, while CPU write accesses have a hold delay of $(MEMHOLD) \times HCLK$ cycles.

To overcome this timing constraint, the attribute memory space can be used by programming its timing register with an ATTHOLD value that meets the t_{WB} timing, and leaving the MEMHOLD value at its minimum. Then, the CPU must use the common memory space for all NAND flash read and write accesses, except when writing the last address byte to the NAND flash device, where the CPU must write to the attribute memory space.

36.6.6 Computation of the error correction code (ECC) in NAND flash memory

The FSMC PC-Card controller includes two error correction code computation hardware blocks, one per memory bank. They are used to reduce the host CPU workload when processing the error correction code by software in the system.

These two registers are identical and associated with bank 2 and bank 3, respectively. As a consequence, no hardware ECC computation is available for memories connected to bank 4.

The error correction code (ECC) algorithm implemented in the FSMC can perform 1-bit error correction and 2-bit error detection per 256, 512, 1 024, 2 048, 4 096 or 8 192 bytes read from or written to NAND flash memory. It is based on the Hamming coding algorithm and consists in calculating the row and column parity.

The ECC modules monitor the NAND flash databus and read/write signals (NCE and NWE) each time the NAND flash memory bank is active.

The functional operations are:

- When access to NAND flash is made to bank 2 or bank 3, the data present on the D[15:0] bus is latched and used for ECC computation.
- When access to NAND flash occurs at any other address, the ECC logic is idle, and does not perform any operation. Thus, write operations for defining commands or addresses to NAND flash are not taken into account for ECC computation.

Once the desired number of bytes has been read from/written to the NAND flash by the host CPU, the FSMC_ECCR2/3 registers must be read in order to retrieve the computed value. Once read, they should be cleared by resetting the ECCEN bit to zero. To compute a new data block, the ECCEN bit must be set to one in the FSMC_PCR2/3 registers.

To perform an ECC computation:

1. Enable the ECCEN bit in the FSMC_PCR2/3 register.
2. Write data to the NAND flash memory page. While the NAND page is written, the ECC block computes the ECC value.
3. Read the ECC value available in the FSMC_ECCR2/3 register and store it in a variable.
4. Clear the ECCEN bit and then enable it in the FSMC_PCR2/3 register before reading back the written data from the NAND page. While the NAND page is read, the ECC block computes the ECC value.
5. Read the new ECC value available in the FSMC_ECCR2/3 register.
6. If the two ECC values are the same, no correction is required, otherwise there is an ECC error and the software correction routine returns information on whether the error can be corrected or not.

36.6.7 PC Card/CompactFlash operations

Address spaces and memory accesses

The FSMC supports Compact Flash storage or PC Cards in Memory Mode and I/O Mode (True IDE mode is not supported).

The Compact Flash storage and PC Cards are made of 3 memory spaces:

- Common Memory Space
- Attribute Space
- I/O Memory Space

The nCE2 and nCE1 pins (FSMC_NCE4_2 and FSMC_NCE4_1 respectively) select the card and indicate whether a byte or a word operation is being performed: nCE2 accesses the odd byte on D15-8 and nCE1 accesses the even byte on D7-0 if A0=0 or the odd byte on D7-0 if A0=1. The full word is accessed on D15-0 if both nCE2 and nCE1 are low.

The memory space is selected by asserting low nOE for read accesses or nWE for write accesses, combined with the low assertion of nCE2/nCE1 and nREG.

- If pin nREG=1 during the memory access, the common memory space is selected
- If pin nREG=0 during the memory access, the attribute memory space is selected

The I/O Space is selected by asserting low nIORD for read accesses or nIOWR for write accesses [instead of nOE/nWE for memory Space], combined with nCE2/nCE1. Note that nREG must also be asserted low during accesses to I/O Space.

Three type of accesses are allowed for a 16-bit PC Card:

- Accesses to Common Memory Space for data storage can be either 8-bit accesses at even addresses or 16 bit AHB accesses.
Note that 8-bit accesses at odd addresses are not supported and do not lead to the low assertion of nCE2. A 32-bit AHB request is translated into two 16-bit memory accesses.
- Accesses to Attribute Memory Space where the PC Card stores configuration information are limited to 8-bit AHB accesses at even addresses.
Note that a 16-bit AHB access is converted into a single 8-bit memory transfer: nCE1 is asserted low, nCE2 is asserted high and only the even Byte on D7-D0 is valid. Instead

a 32-bit AHB access is converted into two 8-bit memory transfers at even addresses: nCE1 is asserted low, NCE2 is asserted high and only the even bytes are valid.

- Accesses to I/O Space can be performed either through AHB 8-bit or 16-bit accesses.

Table 251. 16-bit PC-Card signals and access type

nCE2	nCE1	nREG	nOE/nWE	nIORD /nIOWR	A10	A9	A7-1	A0	Space	Access Type	Allowed/not Allowed
1	0	1	0	1	X	X	X-X	X	Common Memory Space	Read/Write byte on D7-D0	YES
0	1	1	0	1	X	X	X-X	X		Read/Write byte on D15-D8	Not supported
0	0	1	0	1	X	X	X-X	0		Read/Write word on D15-D0	YES
X	0	0	0	1	0	1	X-X	0	Attribute Space	Read or Write Configuration registers	YES
X	0	0	0	1	0	0	X-X	0		Read or Write CIS (Card Information Structure)	YES
1	0	0	0	1	X	X	X-X	1	Attribute Space	Invalid Read or Write (odd address)	YES
0	1	0	0	1	X	X	X-X	x		Invalid Read or Write (odd address)	YES
1	0	0	1	0	X	X	X-X	0	I/O space	Read Even Byte on D7-0	YES
1	0	0	1	0	X	X	X-X	1		Read Odd Byte on D7-0	YES
1	0	0	1	0	X	X	X-X	0		Write Even Byte on D7-0	YES
1	0	0	1	0	X	X	X-X	1		Write Odd Byte on D7-0	YES
0	0	0	1	0	X	X	X-X	0		Read Word on D15-0	YES
0	0	0	1	0	X	X	X-X	0		Write word on D15-0	YES
0	1	0	1	0	X	X	X-X	X		Read Odd Byte on D15-8	Not supported
0	1	0	1	0	X	X	X-X	X		Write Odd Byte on D15-8	Not supported

The FSMC Bank 4 gives access to those 3 memory spaces as described in [Section 36.4.2: NAND/PC Card address mapping](#) and [Table 218: Memory mapping and timing registers](#).

Wait Feature

The CompactFlash Storage or PC Card may request the FSMC to extend the length of the access phase programmed by MEMWAITx/ATTWAITx/IOWAITx bits, asserting the nWAIT signal after nOE/nWE or nIORD/nIOWR activation if the wait feature is enabled through the PWAITEN bit in the FSMC_PCRx register. In order to detect the nWAIT assertion correctly, the MEMWAITx/ATTWAITx/IOWAITx bits must be programmed as follows:

$xxWAITx \geq 4 + \text{max_wait_assertion_time}/\text{HCLK}$

Where max_wait_assertion_time is the maximum time taken by NWAIT to go low once nOE/nWE or nIORD/nIOWR is low.

After the de-assertion of nWAIT, the FSMC extends the WAIT phase for 4 HCLK clock cycles.

36.6.8 NAND flash/PC Card control registers

The NAND flash/PC Card control registers have to be accessed by words (32 bits).

PC Card/NAND flash control registers 2..4 (FSMC_PCR2..4)

Address offset: $0xA0000000 + 0x40 + 0x20 * (x - 1)$, $x = 2..4$

Reset value: 0x0000 0018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												ECCPS[2:0]			TAR[2:0]				TCLR[2:0]				Res.	ECCEN	PWID[1:0]		PTYP	PBKEN	PWAITEN	Reserved	
												r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w		r/w	r/w	r/w	r/w	r/w	r/w		r/w

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:17 **ECCPS[2:0]**: ECC page size.

Defines the page size for the extended ECC:

000: 256 bytes
 001: 512 bytes
 010: 1024 bytes
 011: 2048 bytes
 100: 4096 bytes
 101: 8192 bytes

Bits 16:13 **TAR[2:0]**: ALE to RE delay.

Sets time from ALE low to RE low in number of AHB clock cycles (HCLK).

Time is: $t_{ar} = (TAR + SET + 2) \times \text{THCLK}$ where THCLK is the HCLK clock period

0000: 1 HCLK cycle (default)
 1111: 16 HCLK cycles

Note: SET is MEMSET or ATTSET according to the addressed space.

Bits 12:9 **TCLR[2:0]**: CLE to RE delay.

Sets time from CLE low to RE low in number of AHB clock cycles (HCLK).

Time is: $t_{clr} = (TCLR + SET + 2) \times \text{THCLK}$ where THCLK is the HCLK clock period

0000: 1 HCLK cycle (default)
 1111: 16 HCLK cycles

Note: SET is MEMSET or ATTSET according to the addressed space.

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **ECCEN**: ECC computation logic enable bit

0: ECC logic is disabled and reset (default after reset),
 1: ECC logic is enabled.

Bits 5:4 **PWID[1:0]**: Databus width.

Defines the external memory device width.

00: 8 bits

01: 16 bits (default after reset). This value is mandatory for PC Cards.

10: reserved, do not use

11: reserved, do not use

Bit 3 **PTYP**: Memory type.

Defines the type of device attached to the corresponding memory bank:

0: PC Card, CompactFlash, CF+ or PCMCIA

1: NAND flash (default after reset)

Bit 2 **PBKEN**: PC Card/NAND flash memory bank enable bit.

Enables the memory bank. Accessing a disabled memory bank causes an ERROR on AHB bus

0: Corresponding memory bank is disabled (default after reset)

1: Corresponding memory bank is enabled

Bit 1 **PWAITEN**: Wait feature enable bit.

Enables the Wait feature for the PC Card/NAND flash memory bank:

0: disabled

1: enabled

Note: For a PC Card, when the wait feature is enabled, the MEMWAITx/ATTWAITx/IOWAITx bits must be programmed to a value as follows:

$xxWAITx \geq 4 + \max_wait_assertion_time/HCLK$

Where max_wait_assertion_time is the maximum time taken by NWAIT to go low once nOE/nWE or nLORD/nLOWR is low.

Bit 0 Reserved, must be kept at reset value.

FIFO status and interrupt register 2..4 (FSMC_SR2..4)

Address offset: $0xA000\ 0000 + 0x44 + 0x20 * (x-1)$, $x = 2..4$

Reset value: 0x0000 0040

This register contains information about FIFO status and interrupt. The FSMC has a FIFO that is used when writing to memories to store up to 16 words of data from the AHB.

This is used to quickly write to the AHB and free it for transactions to peripherals other than the FSMC, while the FSMC is draining its FIFO into the memory. This register has one of its bits that indicates the status of the FIFO, for ECC purposes.

The ECC is calculated while the data are written to the memory, so in order to read the correct ECC the software must wait until the FIFO is empty.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																									FE	IF	IL	IR	IFS	ILS	IRS
																									r	rw	rw	rw	rw	rw	rw

Bits 31:7 Reserved, must be kept at reset value.

- Bit 6 **FEMPT**: FIFO empty.
Read-only bit that provides the status of the FIFO
0: FIFO not empty
1: FIFO empty
- Bit 5 **IFEN**: Interrupt falling edge detection enable bit
0: Interrupt falling edge detection request disabled
1: Interrupt falling edge detection request enabled
- Bit 4 **ILEN**: Interrupt high-level detection enable bit
0: Interrupt high-level detection request disabled
1: Interrupt high-level detection request enabled
- Bit 3 **IREN**: Interrupt rising edge detection enable bit
0: Interrupt rising edge detection request disabled
1: Interrupt rising edge detection request enabled
- Bit 2 **IFS**: Interrupt falling edge status
The flag is set by hardware and reset by software.
0: No interrupt falling edge occurred
1: Interrupt falling edge occurred
- Note: This bit is set by programming it to 1 by software.*
- Bit 1 **ILS**: Interrupt high-level status
The flag is set by hardware and reset by software.
0: No Interrupt high-level occurred
1: Interrupt high-level occurred
- Bit 0 **IRS**: Interrupt rising edge status
The flag is set by hardware and reset by software.
0: No interrupt rising edge occurred
1: Interrupt rising edge occurred

Note: This bit is set by programming it to 1 by software.

Common memory space timing register 2..4 (FSMC_PMEM2..4)

Address offset: Address: $0xA000\ 0000 + 0x48 + 0x20 * (x - 1)$, $x = 2..4$

Reset value: 0xFCFC FCFC

Each FSMC_PMEMx ($x = 2..4$) read/write register contains the timing information for PC Card or NAND flash memory bank x, used for access to the common memory space of the 16-bit PC Card/CompactFlash, or to access the NAND flash for command, address write access and data read/write access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEMHIZ[7:0]								MEMHOLD[7:0]								MEMWAIT[7:0]								MEMSET[7:0]							
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	

Bits 31:24 **MEMHIZx[7:0]**: Common memory x databus HiZ time

Defines the number of HCLK clock cycles during which the databus is kept in HiZ after the start of a PC Card/NAND flash write access to common memory space on socket x. Only valid for write transaction:

0000 0000: 1 HCLK cycle
 1111 1110: 255 HCLK cycles
 1111 1111: Reserved

Bits 23:16 **MEMHOLDx[7:0]**: Common memory x hold time

For NAND flash read accesses to the common memory space, these bits define the number of (HCLK+2) clock cycles during which the address is held after the command is deasserted (NWE, NOE).

For NAND flash write accesses to the common memory space, these bits define the number of HCLK clock cycles during which the data are held after the command is deasserted (NWE, NOE).

0000 0000: Reserved
 0000 0001: 1 HCLK cycle for write accesses, 3 HCLK cycles for read accesses
 1111 1110: 254 HCLK cycle for write accesses, 256 HCLK cycles for read accesses
 1111 1111: Reserved

Bits 15:8 **MEMWAITx[7:0]**: Common memory x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (NWE, NOE), for PC Card/NAND flash read or write access to common memory space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: Reserved
 0000 0001: 2 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)
 1111 1110: 255 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)
 1111 1111: Reserved.

Bits 7:0 **MEMSETx[7:0]**: Common memory x setup time

Defines the number of HCLK () clock cycles to set up the address before the command assertion (NWE, NOE), for PC Card/NAND flash read or write access to common memory space on socket x:

0000 0000: 1 HCLK cycle
 1111 1110: 255 HCLK cycles
 1111 1111: Reserved

Attribute memory space timing registers 2..4 (FSMC_PATT2..4)

Address offset: 0xA000 0000 + 0x4C + 0x20 * (x – 1), x = 2..4

Reset value: 0xFCFC FCFC

Each FSMC_PATTx (x = 2..4) read/write register contains the timing information for PC Card/CompactFlash or NAND flash memory bank x. It is used for 8-bit accesses to the attribute memory space of the PC Card/CompactFlash or to access the NAND flash for the last address write access if the timing must differ from that of previous accesses (for Ready/Busy management, refer to [Section 36.6.5: NAND flash prewait functionality](#)).

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ATTHIZ[7:0]								ATTHOLD[7:0]								ATTWAIT[7:0]								ATTSET[7:0]							
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:24 **ATTIHZ[7:0]**: Attribute memory x databus HiZ time

Defines the number of HCLK clock cycles during which the databus is kept in HiZ after the start of a PC CARD/NAND flash write access to attribute memory space on socket x. Only valid for write transaction:

0000 0000: 0 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

Bits 23:16 **ATTHOLD[7:0]**: Attribute memory x hold time

For PC Card/NAND flash read accesses to attribute memory space on socket x, these bits define the number of HCLK clock cycles (HCLK +2) clock cycles during which the address is held after the command is deasserted (NWE, NOE).

For PC Card/NAND flash write accesses to attribute memory space on socket x, these bits define the number of HCLK clock cycles during which the data are held after the command is deasserted (NWE, NOE).

0000 0000: reserved

0000 0001: 1 HCLK cycle for write access, 3 HCLK cycles for read accesses

1111 1110: 254 HCLK cycle for write access, 256 HCLK cycles for read accesses

1111 1111: Reserved

Bits 15:8 **ATTWAIT[7:0]**: Attribute memory x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (NWE, NOE), for PC Card/NAND flash read or write access to attribute memory space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: Reserved

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deassertion of NWAIT)

1111 1111: 255 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)

1111 1111: Reserved.

Bits 7:0 **ATTSET[7:0]**: Attribute memory x setup time

Defines the number of HCLK (+1) clock cycles to set up address before the command assertion (NWE, NOE), for PC CARD/NAND flash read or write access to attribute memory space on socket x:

0000 0000: 1 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

I/O space timing register 4 (FSMC_PIO4)

Address offset: 0xA000 0000 + 0xB0

Reset value: 0xFCFCFCFC

The FSMC_PIO4 read/write registers contain the timing information used to gain access to the I/O space of the 16-bit PC Card/CompactFlash.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IOHI[7:0]								IOHOLD[7:0]								IOWAIT[7:0]								IOSET[7:0]							
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:24 IOHIZ[7:0]: I/O x databus HiZ time

Defines the number of HCLK clock cycles during which the databus is kept in HiZ after the start of a PC Card write access to I/O space on socket x. Only valid for write transaction:

0000 0000: 0 HCLK cycle

1111 1111: 255 HCLK cycles (default value after reset)

Bits 23:16 IOHOLD[7:0]: I/O x hold time

Defines the number of HCLK clock cycles to hold address (and data for write access) after the command deassertion (NWE, NOE), for PC Card read or write access to I/O space on socket x:

0000 0000: reserved

0000 0001: 1 HCLK cycle

1111 1111: 255 HCLK cycles (default value after reset)

Bits 15:8 IOWAIT[7:0]: I/O x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (SMNWE, SMNOE), for PC Card read or write access to I/O space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: reserved, do not use this value

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deassertion of NWAIT)

1111 1111: 256 HCLK cycles (+ wait cycle introduced by the Card deasserting NWAIT) (default value after reset)

Bits 7:0 IOSET[7:0]: I/O x setup time

Defines the number of HCLK (+1) clock cycles to set up the address before the command assertion (NWE, NOE), for PC Card read or write access to I/O space on socket x:

0000 0000: 1 HCLK cycle

1111 1111: 256 HCLK cycles (default value after reset)

ECC result registers 2/3 (FSMC_ECCR2/3)

Address offset: $0xA000\ 0000 + 0x54 + 0x20 * (x - 1)$, $x = 2$ or 3

Reset value: 0x0000 0000

These registers contain the current error correction code value computed by the ECC computation modules of the FSMC controller (one module per NAND flash memory bank). When the CPU reads the data from a NAND flash memory page at the correct address (refer to [Section 36.6.6](#)), the data read from or written to the NAND flash are processed automatically by ECC computation module. At the end of X bytes read (according to the ECCPS field in the FSMC_PCRx registers), the CPU must read the computed ECC value from the FSMC_ECCx registers, and then verify whether these computed parity data are the same as the parity value recorded in the spare area, to determine whether a page is valid, and, to correct it if applicable. The FSMC_ECCRx registers should be cleared after being read by setting the ECCEN bit to zero. For computing a new data block, the ECCEN bit must be set to one.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ECCx[31:0]																															
r																															

Bits 31:0 **ECCx[31:0]**: ECC result

This field provides the value computed by the ECC computation logic. [Table 252](#) hereafter describes the contents of these bit fields.

Table 252. ECC result relevant bits

ECCPS[2:0]	Page size in bytes	ECC bits
000	256	ECC[21:0]
001	512	ECC[23:0]
010	1024	ECC[25:0]
011	2048	ECC[27:0]
100	4096	ECC[29:0]
101	8192	ECC[31:0]

36.6.9 FSMC register map

The following table summarizes the FSMC registers.

Table 253. FSMC register map

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0000	FSMC_BCR1	Reserved												CBURSTRW	CPSIZE[2:0]			ASYNCAWAIT	EXTMOD	WAITEN	WREN	WAITCFG	WRAPMOD	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]		MTYP[0:1]		MUXEN	MBKEN
0008	FSMC_BCR2	Reserved												CBURSTRW	CPSIZE[2:0]			ASYNCAWAIT	EXTMOD	WAITEN	WREN	WAITCFG	WRAPMOD	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]		MTYP[0:1]		MUXEN	MBKEN
0010	FSMC_BCR3	Reserved												CBURSTRW	CPSIZE[2:0]			ASYNCAWAIT	EXTMOD	WAITEN	WREN	WAITCFG	WRAPMOD	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]		MTYP[0:1]		MUXEN	MBKEN
0018	FSMC_BCR4	Reserved												CBURSTRW	CPSIZE[2:0]			ASYNCAWAIT	EXTMOD	WAITEN	WREN	WAITCFG	WRAPMOD	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]		MTYP[0:1]		MUXEN	MBKEN
0004	FSMC_BTR1	Res.	ACCMOD[1:0]		DATLAT[3:0]			CLKDIV[3:0]			BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]									
000C	FSMC_BTR2	Res.	ACCMOD[1:0]		DATLAT[3:0]			CLKDIV[3:0]			BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]									
0014	FSMC_BTR3	Res.	ACCMOD[1:0]		DATLAT[3:0]			CLKDIV[3:0]			BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]									
001C	FSMC_BTR4	Res.	ACCMOD[1:0]		DATLAT[3:0]			CLKDIV[3:0]			BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]									
0104	FSMC_BWTR ₁	Res.	ACC MOD [1:0]		Res.							BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]								
010C	FSMC_BWTR ₂	Res.	ACC MOD [1:0]		Res.							BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]								

Table 253. FSMC register map (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0114	FSMC_BWTR 3	Res.		ACC MOD [1:0]	Res.								BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]							
011C	FSMC_BWTR 4	Res.		ACC MOD [1:0]	Res.								BUSTURN[3:0]			DATAST[7:0]							ADDHLD[3:0]			ADDSET[3:0]							
0xA000 0060	FSMC_PCR2	Reserved												ECCPS[2:0]		TAR[2:0]			TCLR[2:0]			Res.	ECCEN	PWID[1:0]		PTYP	PBKEN	PWAITEN	Reserved				
0xA000 0080	FSMC_PCR3	Reserved												ECCPS[2:0]		TAR[2:0]			TCLR[2:0]			Res.	ECCEN	PWID[1:0]		PTYP	PBKEN	PWAITEN	Reserved				
0xA000 00A0	FSMC_PCR4	Reserved												ECCPS[2:0]		TAR[2:0]			TCLR[2:0]			Res.	ECCEN	PWID[1:0]		PTYP	PBKEN	PWAITEN	Reserved				
0xA000 0064	FSMC_SR2	Reserved																								FEMPT	IFEN	ILEN	IREN	IFS	ILS	IRS	
0xA000 0084	FSMC_SR3	Reserved																								FEMPT	IFEN	ILEN	IREN	IFS	ILS	IRS	
0xA000 00A4	FSMC_SR4	Reserved																								FEMPT	IFEN	ILEN	IREN	IFS	ILS	IRS	
0xA000 0068	FSMC_PMEM 2	MEMHIZ[7:0]				MEMHOLD[7:0]				MEMWAIT[7:0]				MEMSET[7:0]																			
0xA000 0088	FSMC_PMEM 3	MEMHIZ[7:0]				MEMHOLD[7:0]				MEMWAIT[7:0]				MEMSET[7:0]																			
0xA000 00A8	FSMC_PMEM 4	MEMHIZ[7:0]				MEMHOLD[7:0]				MEMWAIT[7:0]				MEMSET[7:0]																			
0xA000 006C	FSMC_PATT2	ATTHIZ[7:0]				ATTHOLD[7:0]				ATTWAIT[7:0]				ATTSET[7:0]																			
0xA000 008C	FSMC_PATT3	ATTHIZ[7:0]				ATTHOLD[7:0]				ATTWAIT[7:0]				ATTSET[7:0]																			
0xA000 00AC	FSMC_PATT4	ATTHIZ[7:0]				ATTHOLD[7:0]				ATTWAIT[7:0]				ATTSET[7:0]																			
0xA000 00B0	FSMC_PIO4	IOHIZ[7:0]				IOHOLD[7:0]				IOWAIT[7:0]				IOSET[7:0]																			
0xA000 0074	FSMC_ECCR2	ECC[31:0]																															
0xA000 0094	FSMC_ECCR3	ECC[31:0]																															

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.